

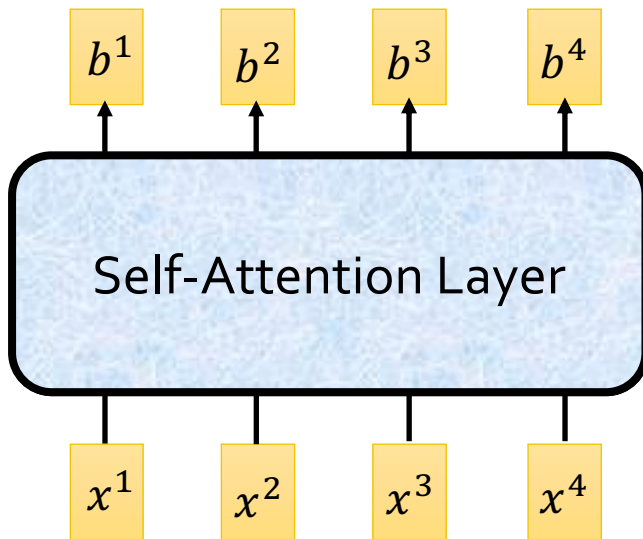
Transformer Architecture

CSCI 601-771 (NLP: Self-Supervised Models)

Self-Attention Module

Self-Attention a dimension-preserving function

- Given a set of input vector, compute output vectors of the same dimension.
- Now what's inside this box?



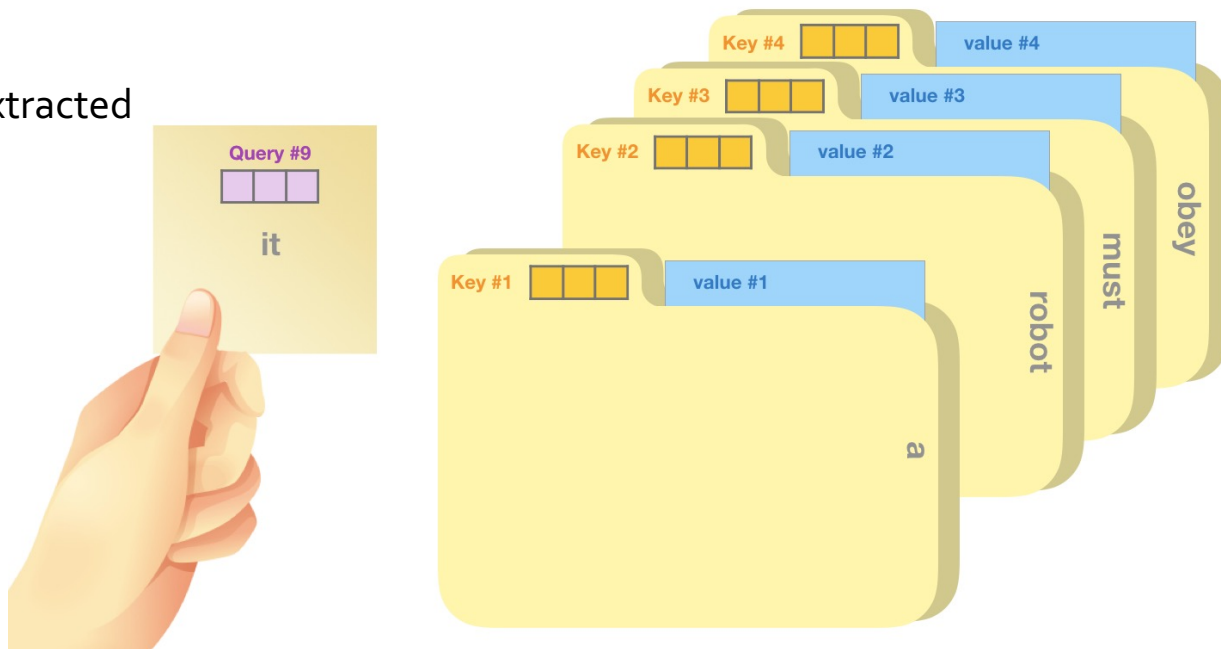
Defining Self-Attention

- Terminology:
 - **Query**: to match others
 - **Key**: to be matched
 - **Value**: information to be extracted

Defining Self-Attention

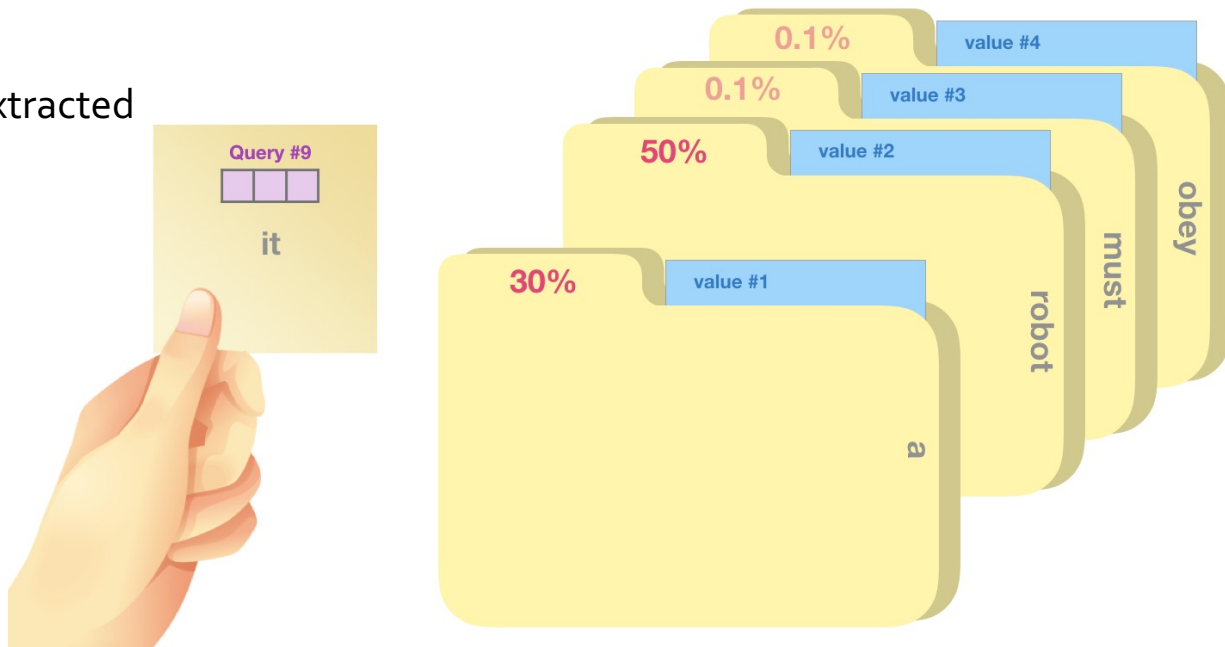
An analogy

- Terminology:
 - **Query**: to match others
 - **Key**: to be matched
 - **Value**: information to be extracted



Defining Self-Attention

- Terminology:
 - **Query**: to match others
 - **Key**: to be matched
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q : query (to match others)

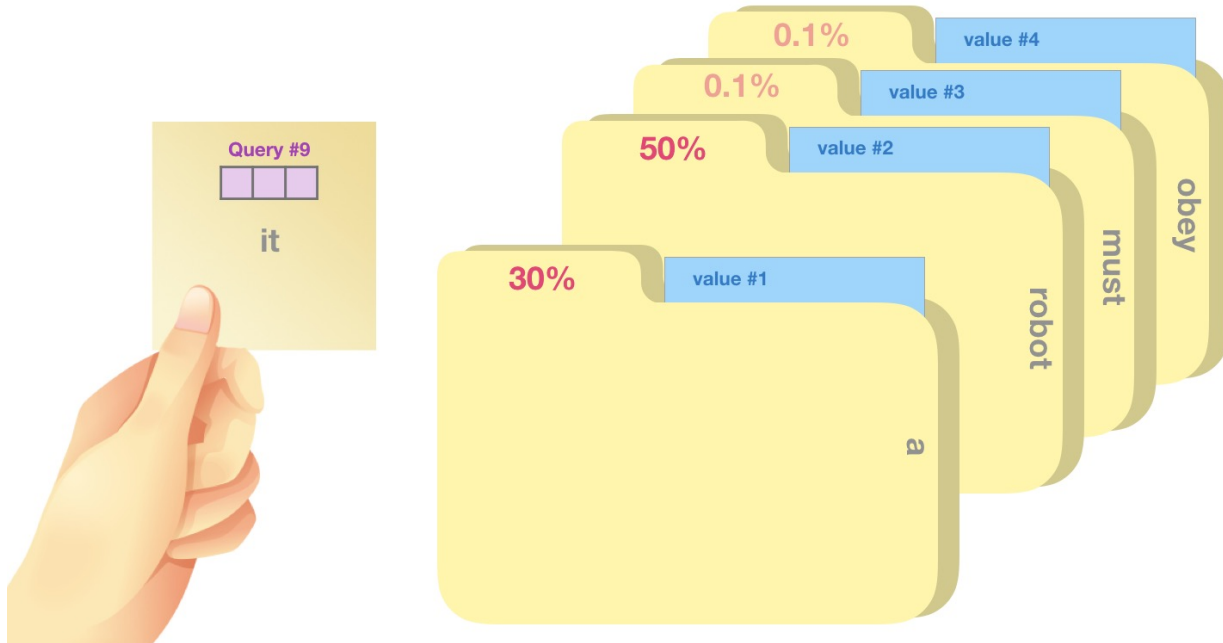
$$q_i = x_i W^q$$

k : key (to be matched)

$$k_i = x_i W^k$$

v : value (information to be extracted)

$$v_i = x_i W^v$$



q : query (to match others)

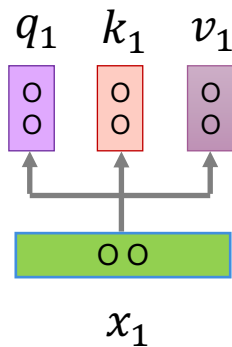
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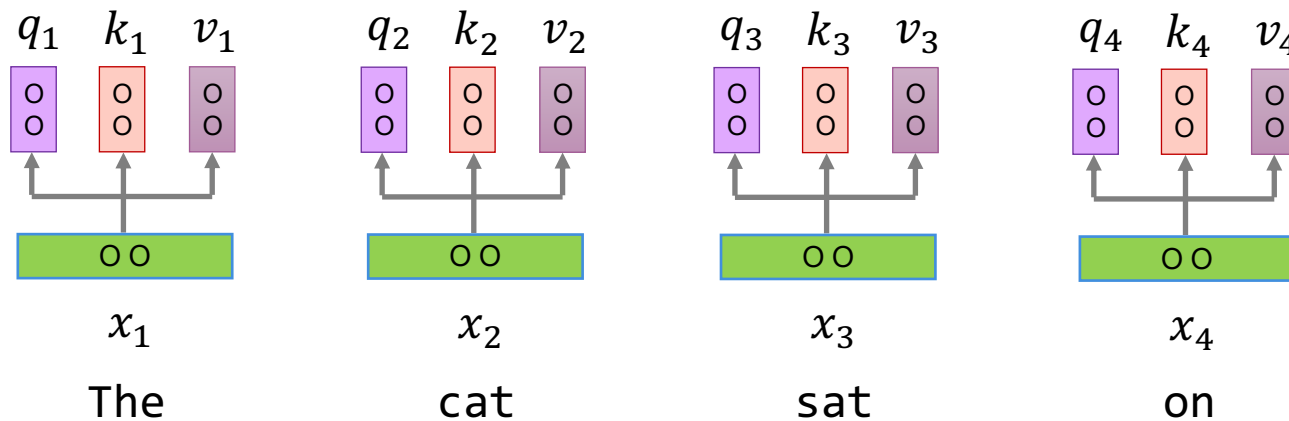
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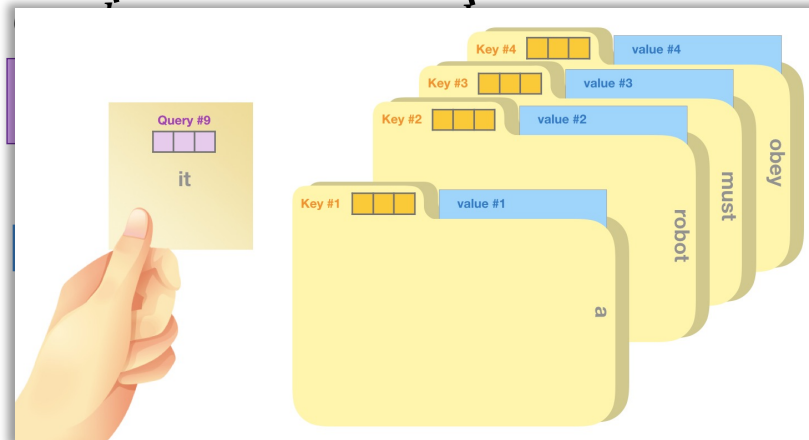
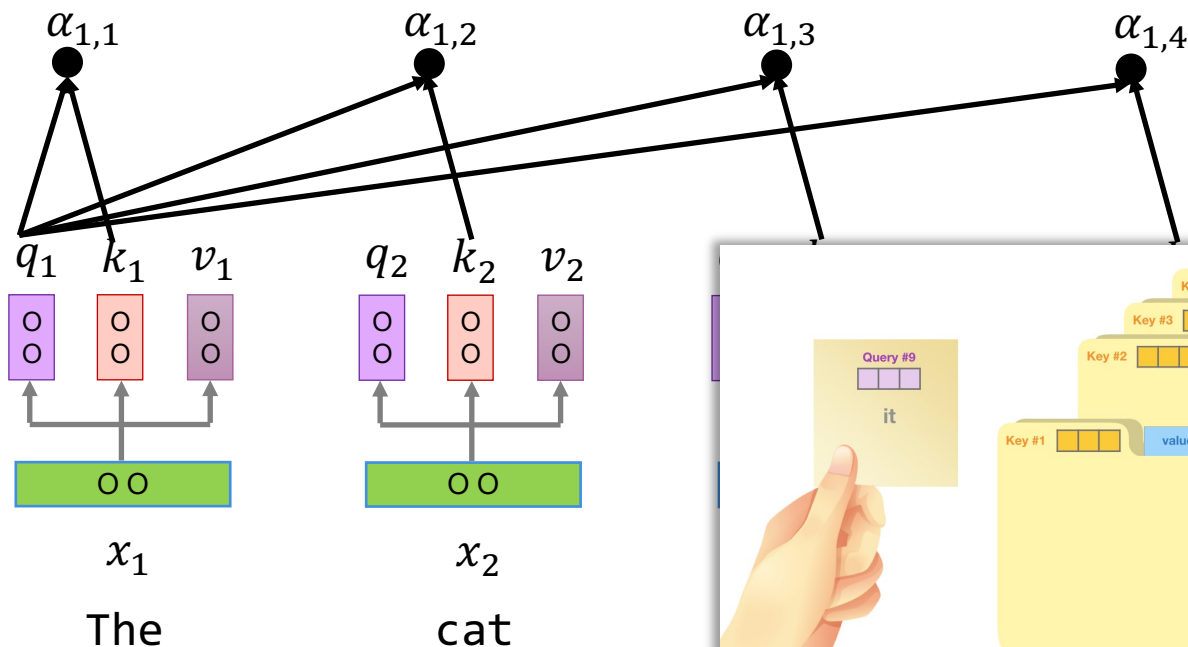
$$\alpha_{1,i} = \underbrace{q^1 \cdot k^i}_{\text{Scaled dot product}} / \sqrt{d}$$

q : query (to match others)

k : key (to be matched)

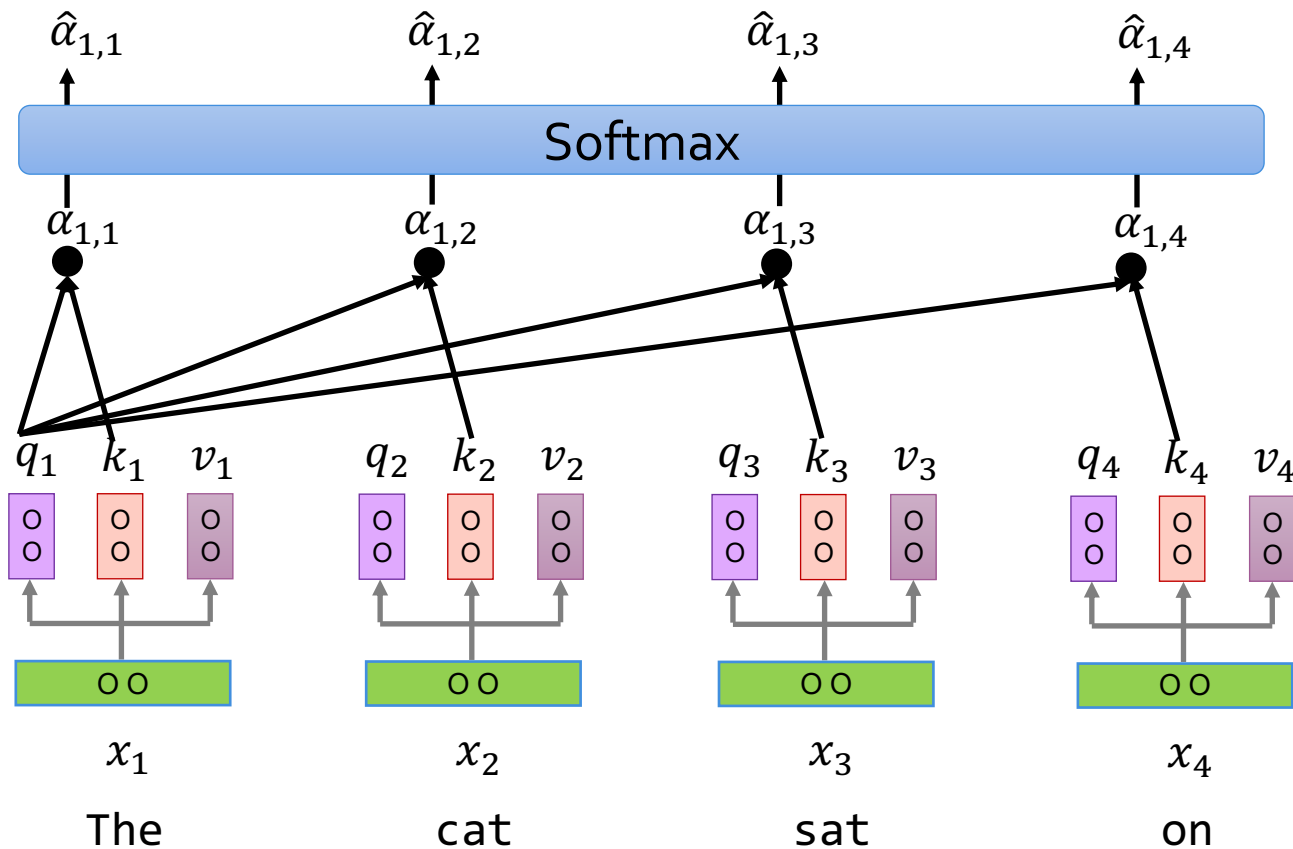
v : value (information to be extracted)

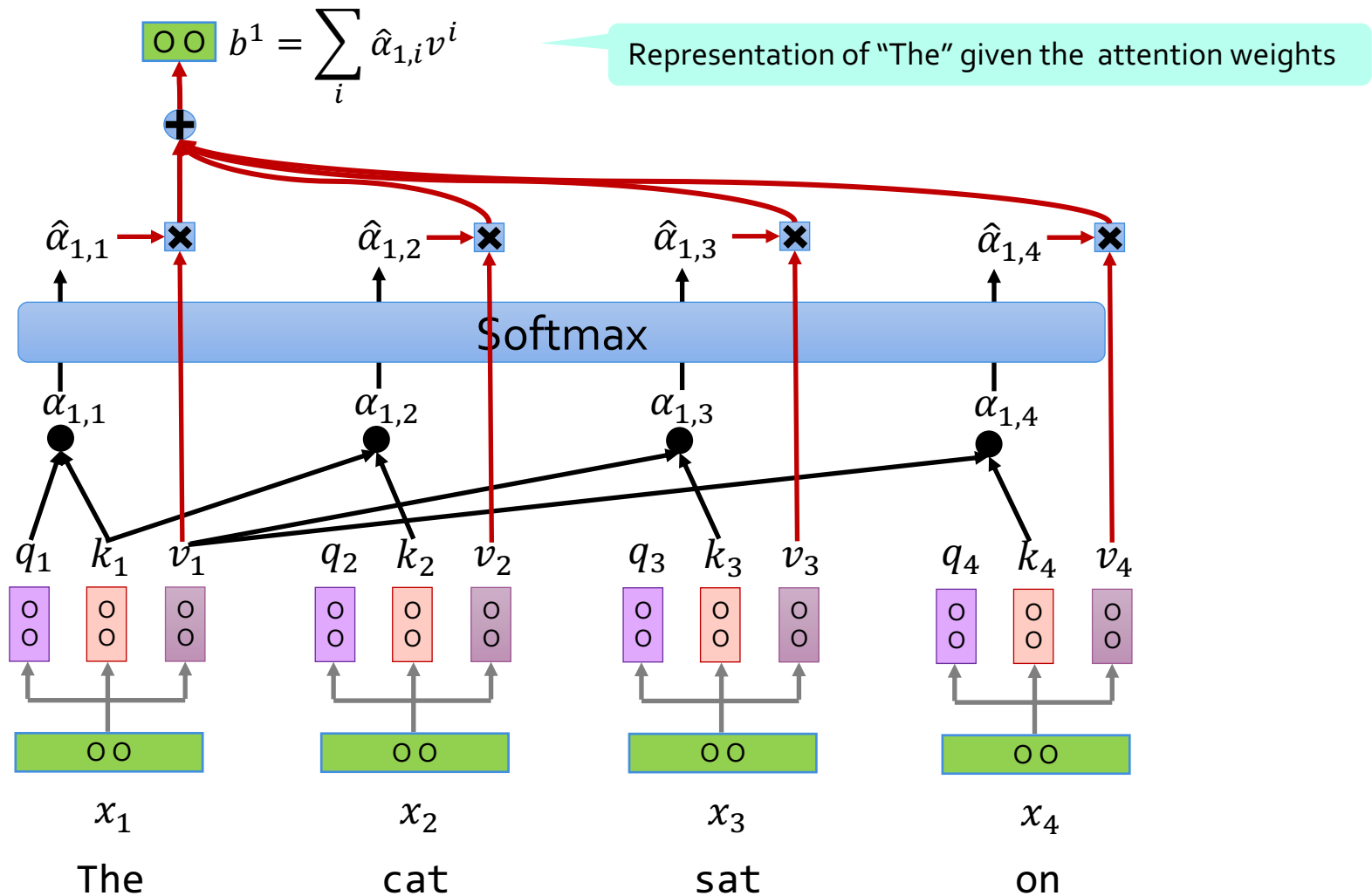
How much should "The" attend to other positions?



$$\sigma(z)_i = \frac{\exp(z_i)}{\sum_j \exp(z_j)}$$

How much
should "The"
attend to other
positions?





Self-Attention: Matrix Notation

- Input sequence: $\mathbf{x} = [x_1, x_2, \dots, x_n] \in \mathbb{R}^{n \times d}$

- Calculate:

$$\mathbf{Q} = \mathbf{x}\mathbf{W}^q \in \mathbb{R}^{n \times d}$$

$$\mathbf{K} = \mathbf{x}\mathbf{W}^k \in \mathbb{R}^{n \times d}$$

$$\mathbf{V} = \mathbf{x}\mathbf{W}^v \in \mathbb{R}^{n \times d}$$

- Pairwise similarity matrix between queries and keys: $\mathbf{Q}\mathbf{K}^T \in \mathbb{R}^{n \times n}$
- Attention output:

$$\text{Attention}(\mathbf{x}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right)\mathbf{V} \in \mathbb{R}^{n \times d}$$

Attention raw scores

0	-0.08	1.24	0.69	-0.98	1.43	-0.6	0.7	0.16	0.93	1.28	-1.61	-1.1
1	-0.09	-0.0	-0.7	0.06	0.25	0.23	0.26	0.18	0.78	-0.21	-1.01	1.01
2	0.86	1.19	1.59	0.86	-0.13	-0.15	-2.13	-0.98	-0.87	-1.72	1.87	-0.72
3	0.12	-0.03	-0.02	0.88	-0.46	-0.7	0.54	-0.42	-1.89	-0.38	0.04	-0.84
4	0.51	0.17	0.13	-1.64	0.24	-0.02	1.68	-0.36	0.64	0.36	0.27	0.66
5	0.24	-1.44	0.43	0.74	0.96	-1.21	-0.31	1.54	1.66	1.14	0.58	-1.44
6	0.26	-0.1	0.93	0.72	-0.38	1.65	0.47	-0.96	-0.17	-0.9	-1.57	0.22
7	-0.55	0.81	0.71	1.7	-0.8	-1.14	-0.32	1.78	-0.7	-0.04	1.54	0.81
8	0.74	-0.76	-0.44	-0.08	-1.38	-0.13	1.25	-1.37	1.84	0.3	0.57	0.74
9	-0.97	-0.91	0.15	0.35	-0.81	0.11	1.14	-1.52	1.06	1.87	0.5	-0.3
10	1.56	0.9	0.39	1.46	1.44	-1.05	0.9	-0.73	0.36	-0.67	-0.62	-0.43
11	0.32	0.74	0.44	-0.1	1.19	0.83	0.29	2.06	0.51	-0.26	1.51	0.11
	1	2	3	4	5	6	7	8	9	10	11	12

$$\begin{array}{c}
 \begin{array}{cc} \mathbf{Q} & \mathbf{K}^T \end{array} \\
 \begin{array}{c} \text{softmax}\left(\frac{\begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} \times \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array}}{\sqrt{d_k}}\right) \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} \end{array} \\
 \mathbf{V} \\
 \begin{array}{c} \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} \\ \hline \end{array} \\
 \mathbf{H}
 \end{array}$$

Q: Why do we need $\sqrt{d}$?

$$\text{Attention}(\mathbf{x}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right)\mathbf{V}$$

- Without $\sqrt{d}$, the results of $\mathbf{Q}\mathbf{K}^T$ can be very large.
- Large differences between scores make softmax nearly one-hot
 - And consequently, small gradients during backprop.
- A theoretical justification: one can prove that:
 - if each element of $\mathbf{Q}$ and $\mathbf{K}$ are drawn independently from a $N(0, \sigma^2)$, then $\mathbf{Q}\mathbf{K}^T$ would have a variance $d\sigma^4$.
 - But $\mathbf{Q}\mathbf{K}^T/\sqrt{d}$ has variance of σ^4 (i.e., **independent of d**).
- This scaling prevents score variance from growing with dimension.

Q: Does this also handle batching?

$$\text{Attention}(\mathbf{x}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right)\mathbf{V}$$

Self-Attention: Thinking about batching

- Suppose our data comes in batches of size b , i.e., $\mathbf{x} \in \mathbb{R}^{b \times n \times d}$
- Self-attention's matrix form extends to this batched data. Let's verify:
- Calculate the keys, queries and values, per-batch:

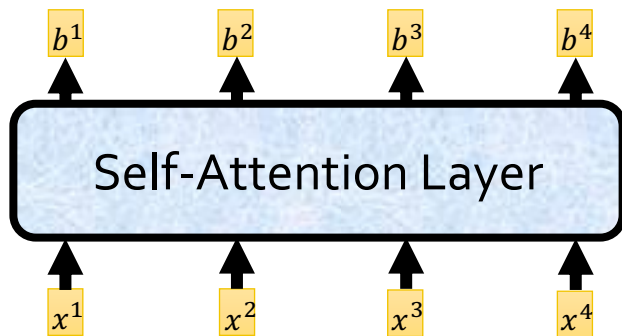
$$\begin{aligned}\mathbf{Q} &= \mathbf{x}\mathbf{W}^q \in \mathbb{R}^{b \times n \times d} \\ \mathbf{K} &= \mathbf{x}\mathbf{W}^k \in \mathbb{R}^{b \times n \times d} \\ \mathbf{V} &= \mathbf{x}\mathbf{W}^v \in \mathbb{R}^{b \times n \times d}\end{aligned}$$

- Pairwise similarity matrix between queries and keys, per batch: $\mathbf{Q}\mathbf{K}^T \in \mathbb{R}^{b \times n \times n}$
- Attention output for the whole batched data:

$$\text{Attention}(\mathbf{x}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d}}\right)\mathbf{V} \in \mathbb{R}^{b \times n \times d}$$

Recap: Self-Attention

- **Attention** is a powerful mechanism to create context-aware representations
- A way to focus on select parts of the input

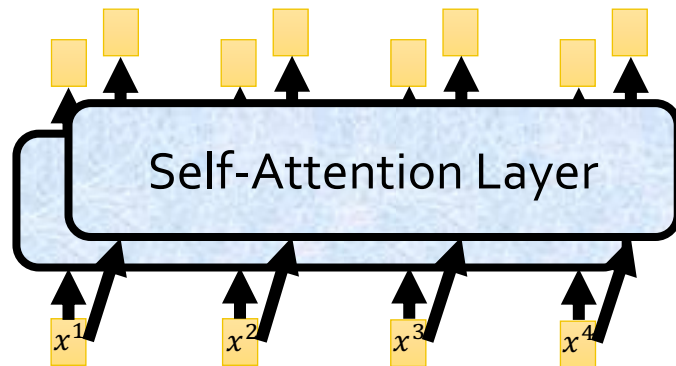


- Better at maintaining **long-distance dependencies** in the context.

Multi-Headed Self-Attention



- **Issue:** Each head has limited capacity.
 - A single head produces one weighted mixture per query; **multiple** heads can capture different relationships.
- **Main idea:** Allows model to jointly attend to information from different representation subspaces (like ensembling).
- **Multiple parallel attention layers.**
 - Each attention layer has its own parameters.
 - Concatenate the results and run them through a linear projection.

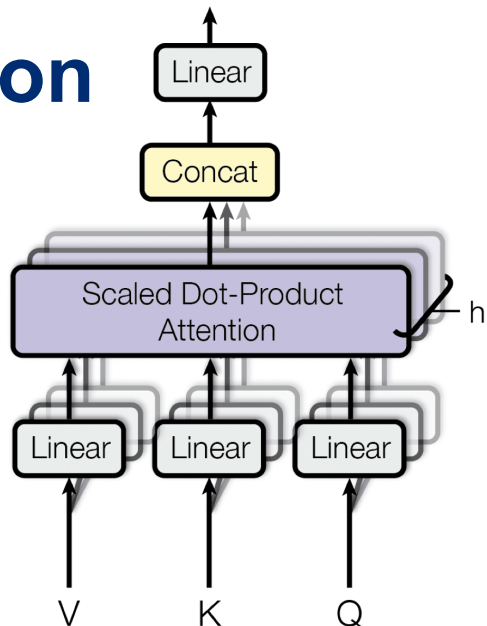


Multi-Headed Self-Attention

- Just concatenate all the heads and apply an output projection.

$$\text{head}_i = \text{Attention}(\mathbf{x}\mathbf{W}_i^q, \mathbf{x}\mathbf{W}_i^k, \mathbf{x}\mathbf{W}_i^v)$$
$$\text{MultiHeadedAttention}(\mathbf{x}) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)\mathbf{W}^o$$

- Q:** How do we change the dimensions of these parameters so that SA is dimension-preserving?



h : number of heads
 d : feature dimension
in output of SA

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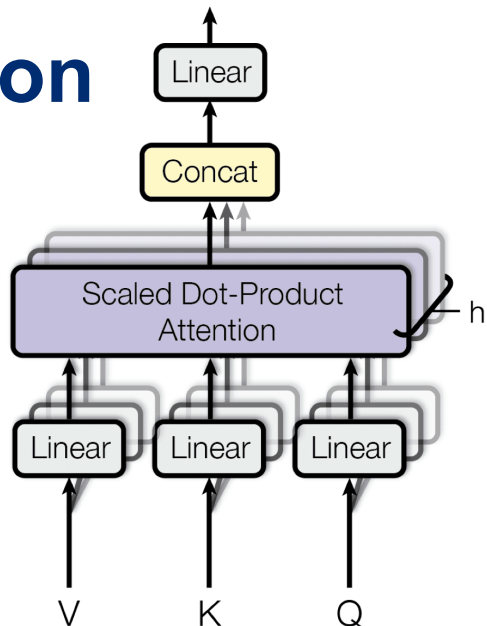
- Previously, we used the following dimensions for **single**-head SA:

$$\mathbf{W}_i^q \in \mathbb{R}^{d \times d}, \quad \mathbf{W}_i^k \in \mathbb{R}^{d \times d}, \quad \mathbf{W}_i^v \in \mathbb{R}^{d \times d},$$

- In practice, we use a reduced dimension for each head.

$$\mathbf{W}_i^q \in \mathbb{R}^{d \times \frac{d}{h}}, \quad \mathbf{W}_i^k \in \mathbb{R}^{d \times \frac{d}{h}}, \quad \mathbf{W}_i^v \in \mathbb{R}^{d \times \frac{d}{h}}, \quad \mathbf{W}^o \in \mathbb{R}^{d \times d}$$

- The total computational cost is similar to that of single-head attention with full dimensionality.



h : number of heads
 d : feature dimension
in output of SA

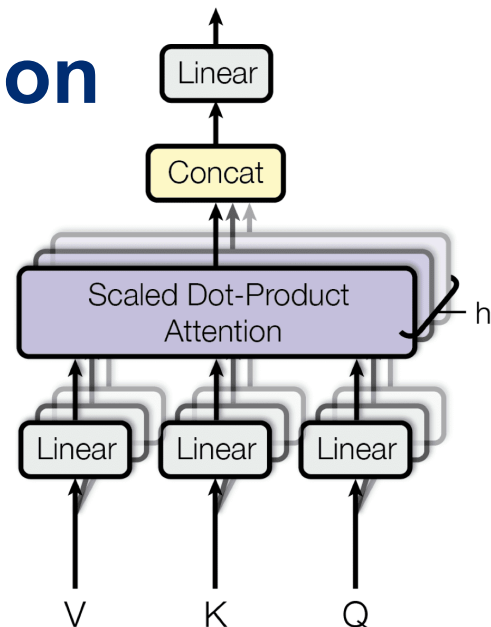
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- Pulse check:**

- **Q1:** What is the rank of each head's output?
 - **A1:** each head's outputs has rank at most d/h .
 - This is because the information pass through each head's rank low-dimensional bottleneck.
 - i.e., each head can only write within a d/h -dim subspace.
- **Q2:** What is the rank of the overall output of MHA?
- **A2:** Note the combined output (after concat and $\mathbf{W}^o$) can have rank up to $\min(n, d)$, where n is the sequence length.



h : number of heads
 d : feature dimension
in output of SA

Implementing Self-Attention Efficiently

- Modern LLMs don't use loops or separate W matrices. They instead use a fused QKV matrix.

```
def mha_textbook(x):  
    heads = []  
    for i in range(h):  
        q, k, v = x @ Wq[i], x @ Wk[i], x @ Wv[i]      # each (B, T, dh)  
        scores = q @ k.transpose(-2, -1) / math.sqrt(dh)  # (B, T, T)  
        scores = scores.masked_fill(mask, float("-inf"))  
        heads.append(torch.softmax(scores, dim=-1) @ v)    # (B, T, dh)  
    return torch.cat(heads, dim=-1) @ Wo                 # (B, T, d)
```

Given the fused weight ->

```
W_qkv = torch.cat([torch.cat(Wq, dim=1),      # (d, d)
                    torch.cat(Wk, dim=1),
                    torch.cat(Wv, dim=1)], dim=1)      # (d, 3d)
```

```
def mha_fused(x):
```

```
    qkv = x @ W_qkv
```

```
    # Split q/k/v and expose the head axis:
```

```
    # (B, T, 3d) -> (B, T, 3, h, dh) -> (3, B, h, T, dh)
```

```
    qkv = qkv.view(B, T, 3, h, dh).permute(2, 0, 3, 1, 4)
```

```
    q, k, v = qkv.unbind(0)      # each (B, h, T, dh)
```

```
    # Batched over (B, h) -- the loop is gone, not reimplemented.
```

```
    scores = q @ k.transpose(-2, -1) / math.sqrt(dh)      # (B, h, T, T)
```

```
    scores = scores.masked_fill(mask, float("-inf"))
```

```
    ctx = torch.softmax(scores, dim=-1) @ v      # (B, h, T, dh)
```

```
    # The "Concat" box: undo the head split. Reshape only, zero FLOPs.
```

```
    ctx = ctx.transpose(1, 2).reshape(B, T, d)
```

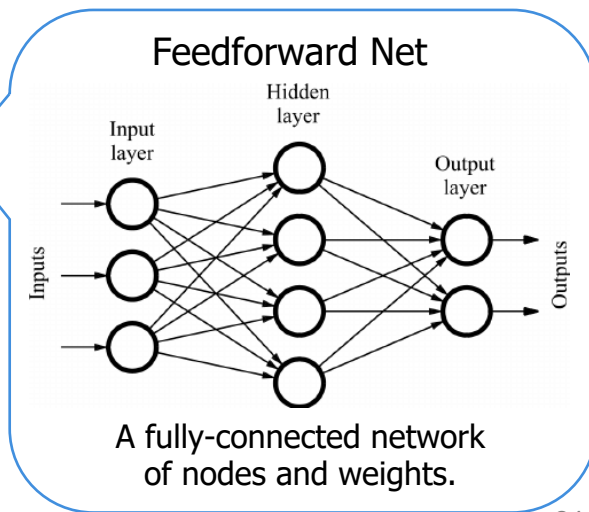
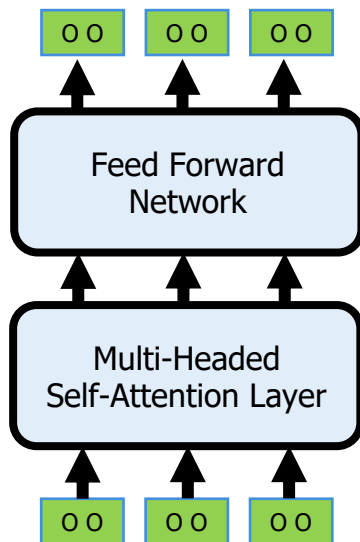
```
    return ctx @ Wo      # ONE GEMM
```

Combine with FFN

- Add a **feed-forward network** to add more expressivity.
 - This applies another nonlinearity to the representations (or “post-process” them).
 - The same FFN is applied **independently** at every token position (unlike attention that mixes information between different positions).

$$\text{FFN}(\mathbf{x}) = f(\mathbf{x}\mathbf{W}_1 + b_1)\mathbf{W}_2 + b_2$$
$$\mathbf{W}_1 \in \mathbb{R}^{d \times d_{\text{ff}}},$$
$$\mathbf{W}_2 \in \mathbb{R}^{d_{\text{ff}} \times d}$$

- Usually, the dimensionality of the hidden feedforward layer d_{ff} is 2-8 times larger than the input dimension d .

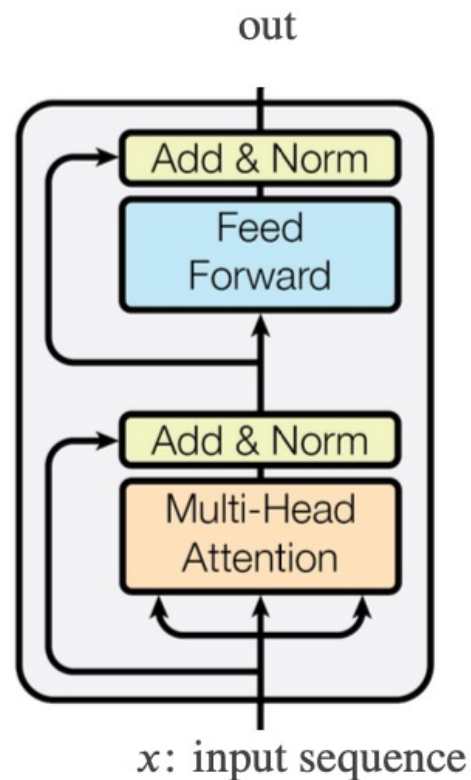


How Do We Prevent Vanishing Gradients?

- **Residual connections** let the model “skip” layers
 - These connections are particularly useful for training deep networks.

$$y = \frac{x - \mathbf{E}[x]}{\sqrt{\mathbf{Var}[x] + \epsilon}} * \gamma + \beta$$

- Use **layer normalization** to stabilize the network and allow for proper gradient flow.
- The visualized variant is called **post-norm**. Nowadays is more common to have them as **pre-norm**. (more on this later).



Transformer Block



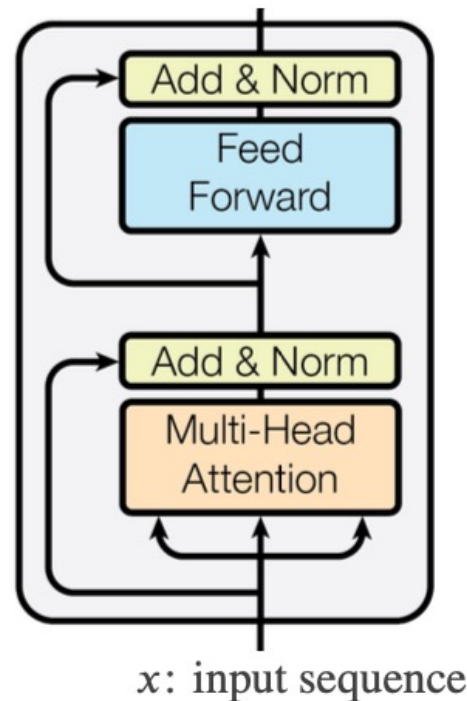
Given input $\mathbf{x}$:

- $\tilde{\mathbf{x}} = \text{MultiHeadedAttention}(\mathbf{x}; W^q, W^k, W^v, W^o)$

$$\mathbf{x} = \text{LayerNorm}(\tilde{\mathbf{x}} + \mathbf{x})$$

$$\tilde{\mathbf{x}} = \text{FFN}(\mathbf{x})$$

$$\text{out} = \text{LayerNorm}(\tilde{\mathbf{x}} + \mathbf{x})$$



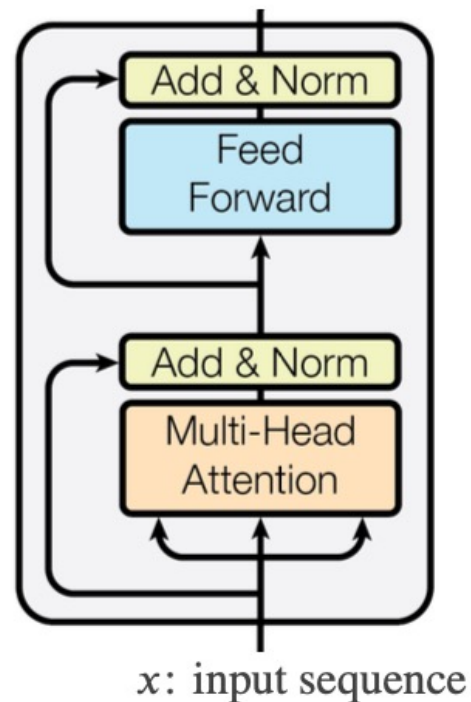
Pulse check

- **Q:** Each attention head uses one set of Q/K/V projection matrices shared across all token positions.
- **A: True.** Projections are shared across positions within a head.
- **Q:** Attention weights across queries sum to 1.
- **A: False.** Softmax is applied across **key positions**, separately for each query.

Summary: Transformer Block



- **Self-Attention:** A critical building block of modern language models.
 - The idea is to compose meanings of words weighted according some similarity notion.
- **Next:** We will extend this block to build various architectures known as Transformer.

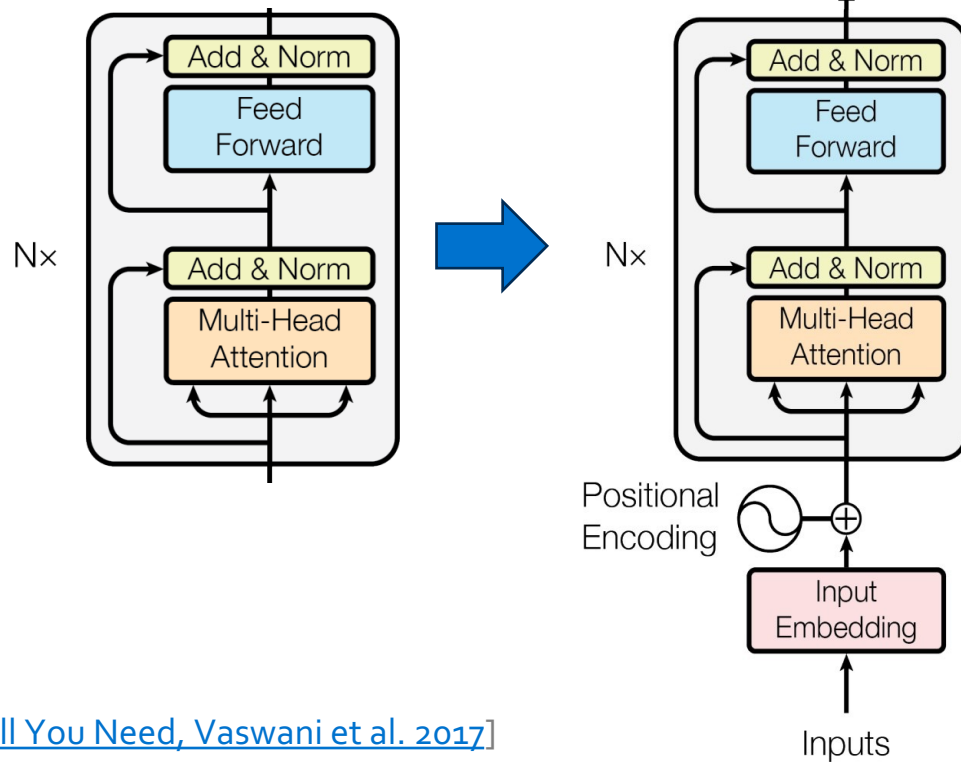
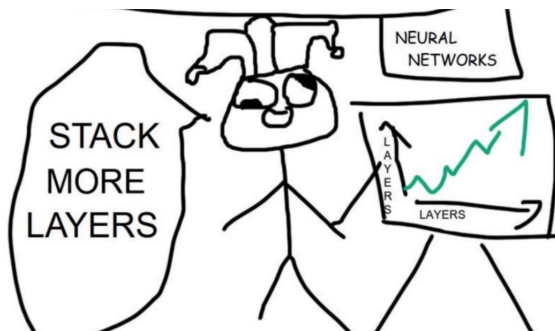




A Decoder-Only Transformer

How Do We Make it Deep?

- Stack more layers!



From Representations to Prediction

- To perform prediction, add a classification head on top of the final layer of the transformer: $\mathbf{x} \in \mathbb{R}^{n \times d}$.
- To obtain logits, we can apply a linear transformation with token embedding matrix $\mathbf{W}_V \in \mathbb{R}^{V \times d}$:

$$\mathbf{x}\mathbf{W}_V^T \in \mathbb{R}^{n \times V}$$

- To obtain probabilities, run this through softmax:

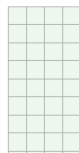
$$\text{softmax}(\mathbf{x}\mathbf{W}_V^T) \in \mathbb{R}^{n \times V}$$

Decoder #12, Position #1
output vector



X

Token
Embeddings



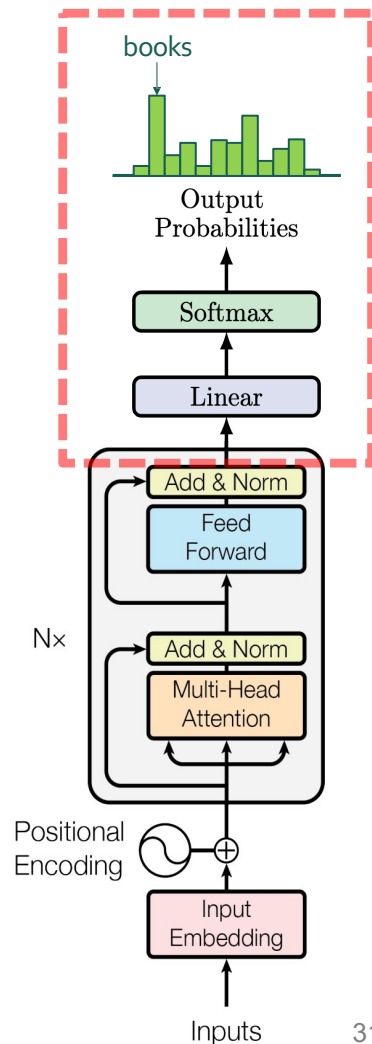
=

output token
probabilities (logits)

0.19850038	aardvark
0.7089803	aarhus
0.46333563	aaron
...	...
...	...
...	...
...	...

Pick an output
token based on
its probability
(sample)

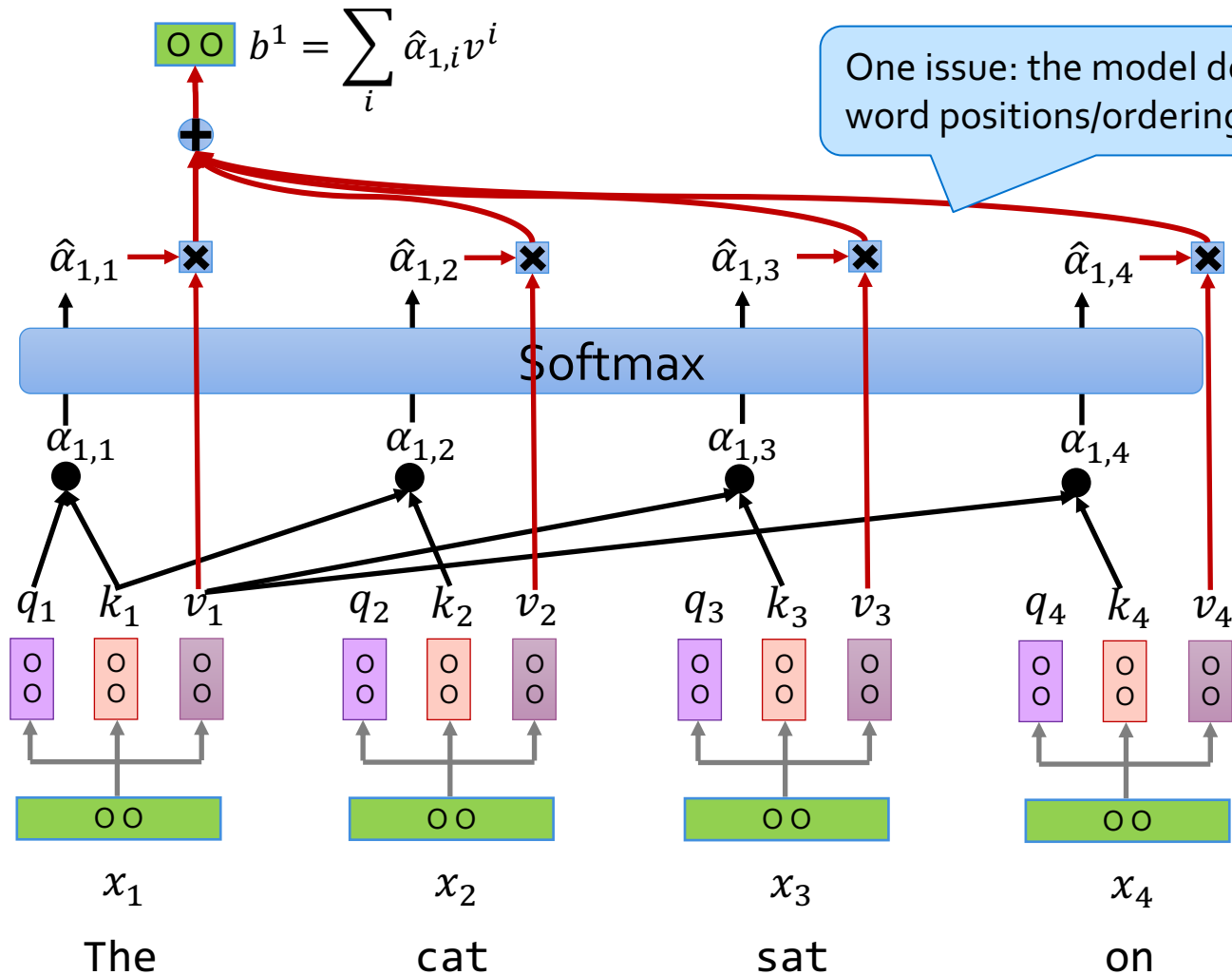
The





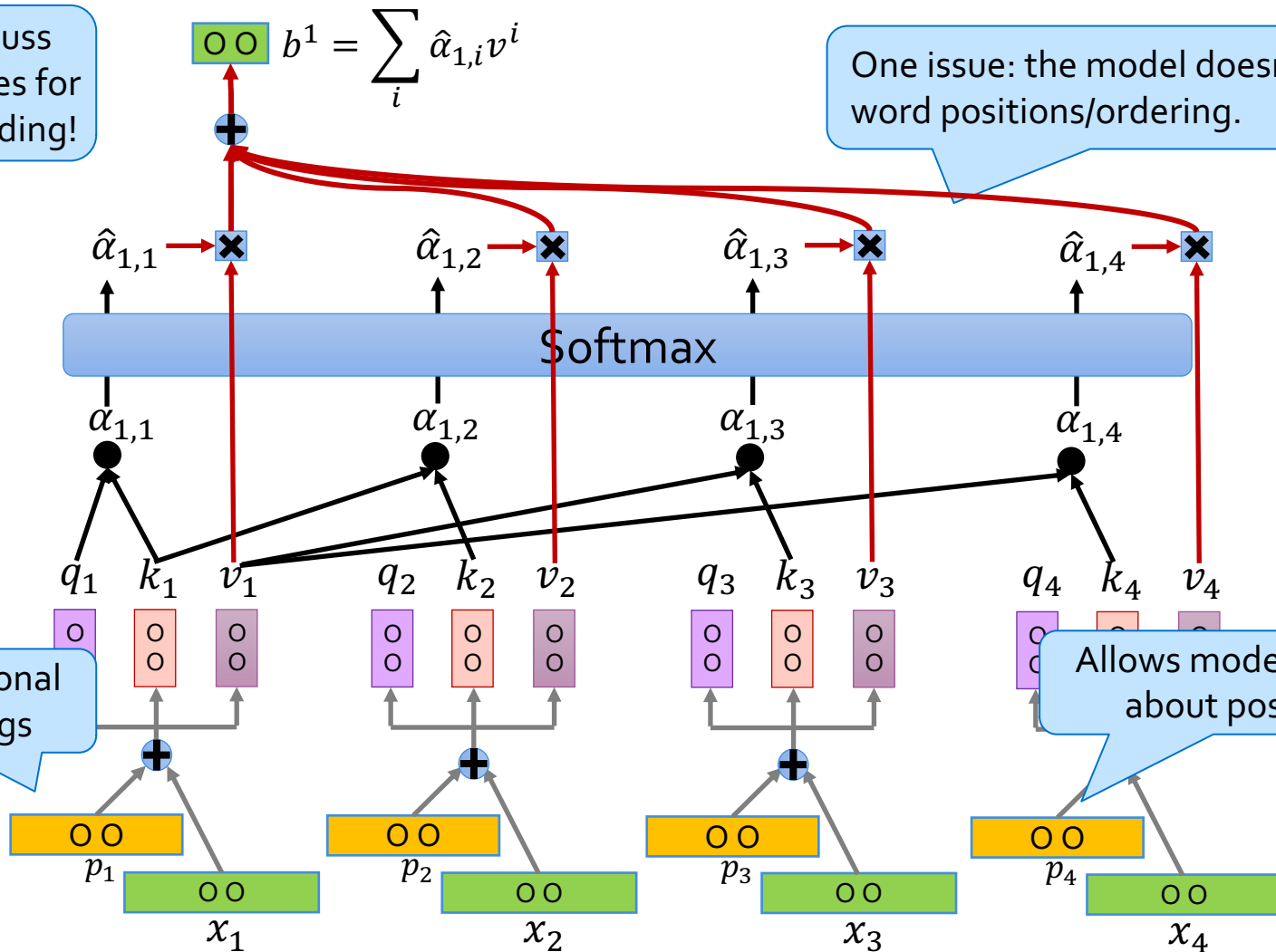
One wrinkle: How does Transformer
know about word ordering? ...





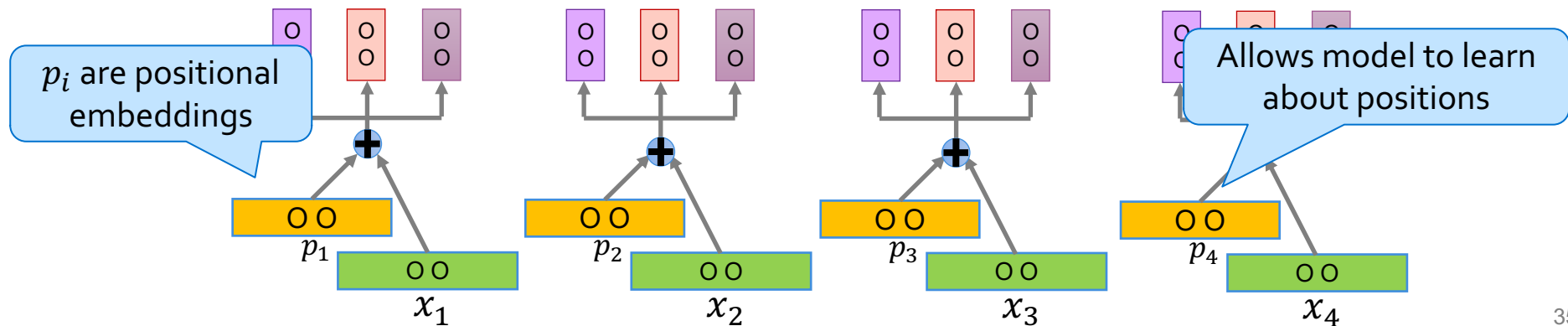
We will discuss various choices for these embedding!

One issue: the model doesn't know word positions/ordering.



Absolute Positional Embeddings

- Add unique vectors for each position.
- They inform Transformer about word orderings.
- We will see more variations of these in later class.

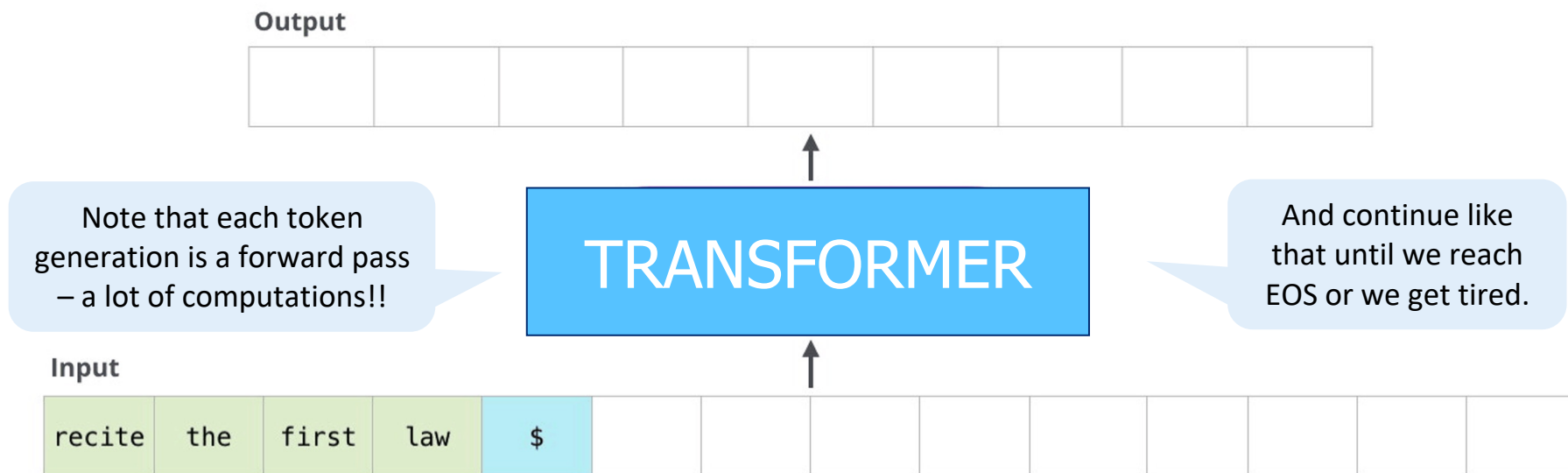




How does inference work?

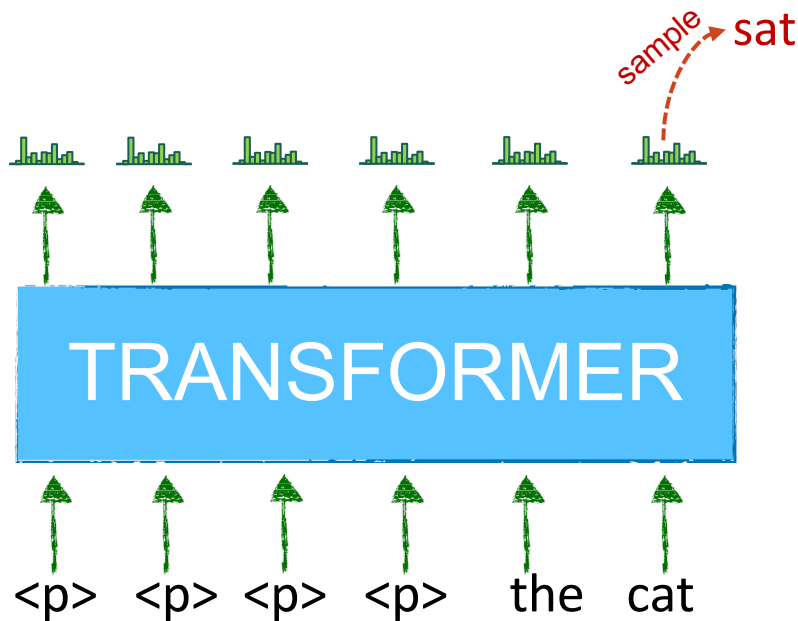


Generating text via Transformer



Generating text via Transformer

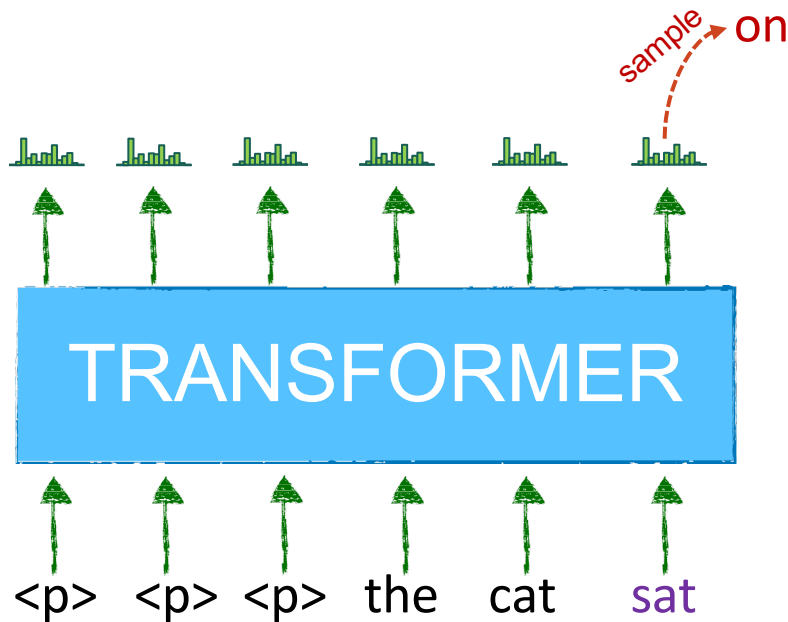
- Use the output of previous step as input to the next step repeatedly



Generating text via Transformer

- Use the output of previous step as input to the next step repeatedly

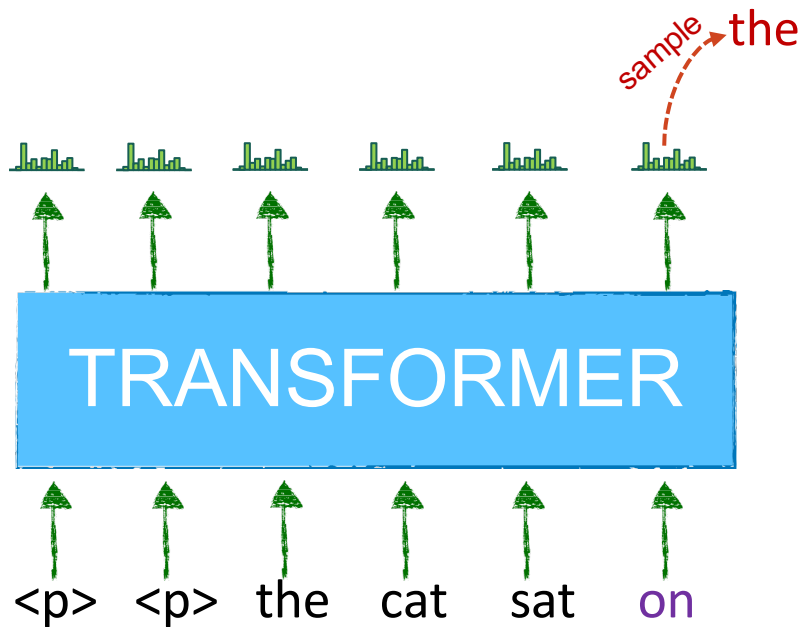
The probabilities get revised upon adding a new token to the input.



Generating text via Transformer

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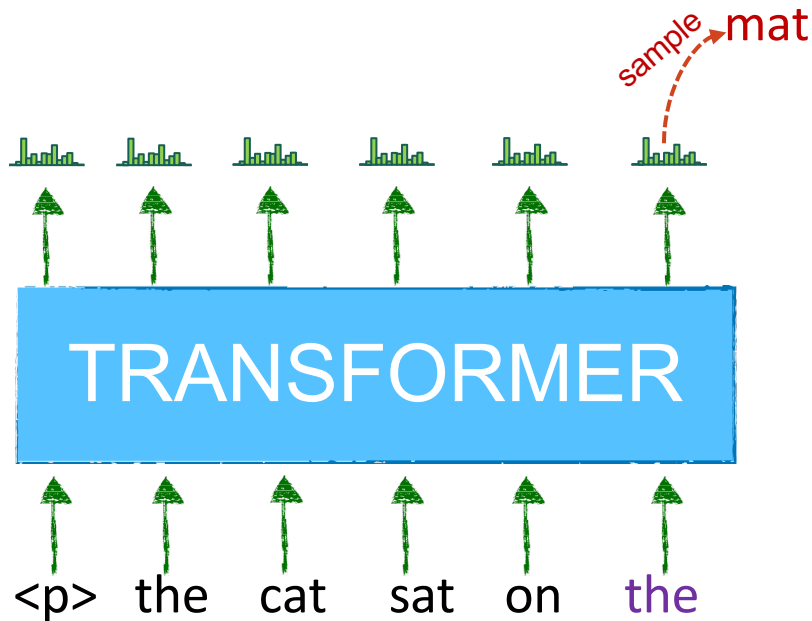
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Generating text via Transformer

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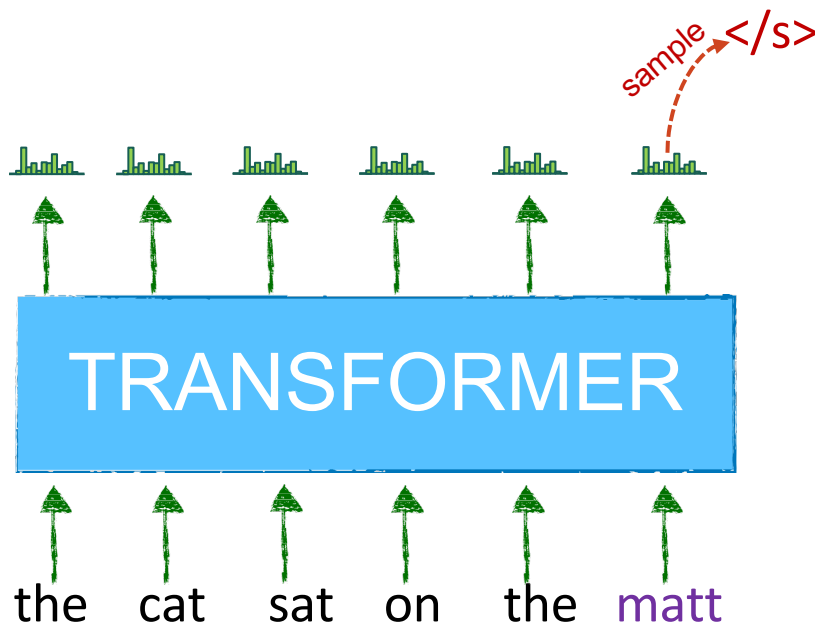
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Generating text via Transformer

- Use the output of previous step as input to the next step repeatedly

The probabilities get revised upon adding a new token to the input.





How does training work?



Quiz

- What do we optimize when we train a transformer?
 1. X and q
 2. W_k and W_v
 3. Softmax function
 4. Sine/cos positional encodings
 5. The embedding matrix
 6. W_1 and W_2 in your FFN
 7. W_q

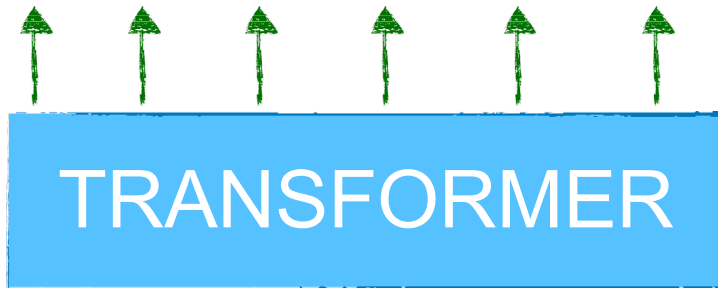
Answer: 2, 6, 7 and 5. You may also be able to train certain positional encodings.

Training a Transformer Language Model

- **Goal:** Train a Transformer for language modeling (i.e., predicting the next word).
- **Approach:** Train it so that each position is predictor of the next (right) token.
 - We just shift the input to left by one, and use as labels

(gold output) $Y =$ cat sat on the mat $</s>$

EOS special token



$X =$ the cat sat on the mat

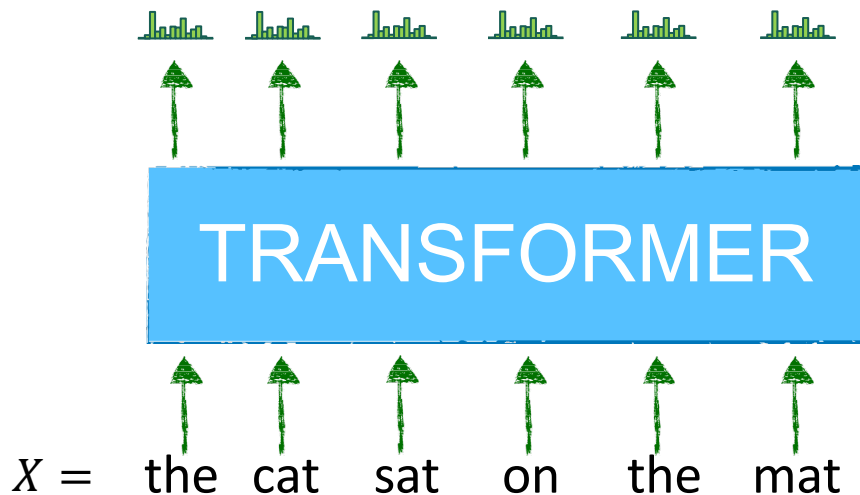
```
X = text[:, :-1]
Y = text[:, 1:]
```

[Slide credit: Arman Cohan]

Training a Transformer Language Model

- For each position, compute their corresponding **distribution** over the whole vocab.

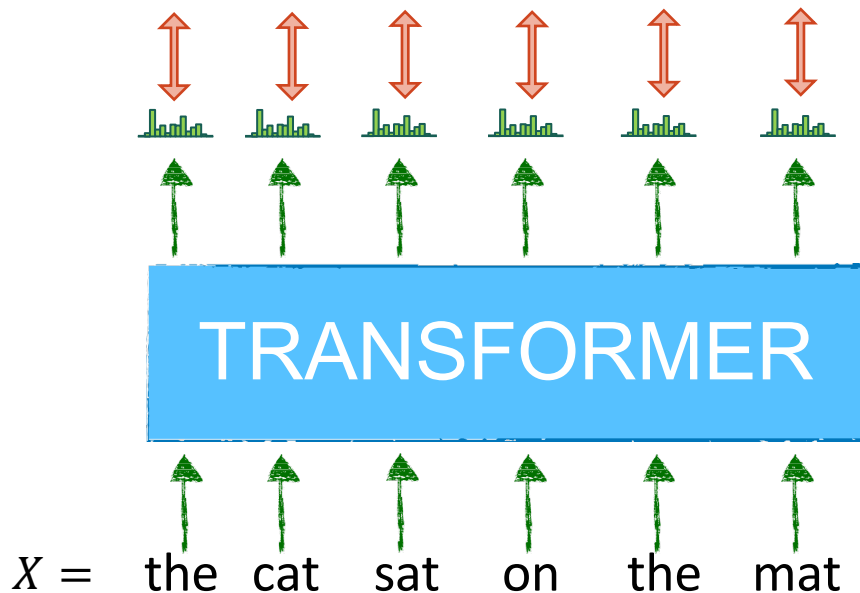
(gold output) $Y =$ cat sat on the mat $\langle /s \rangle$



Training a Transformer Language Model

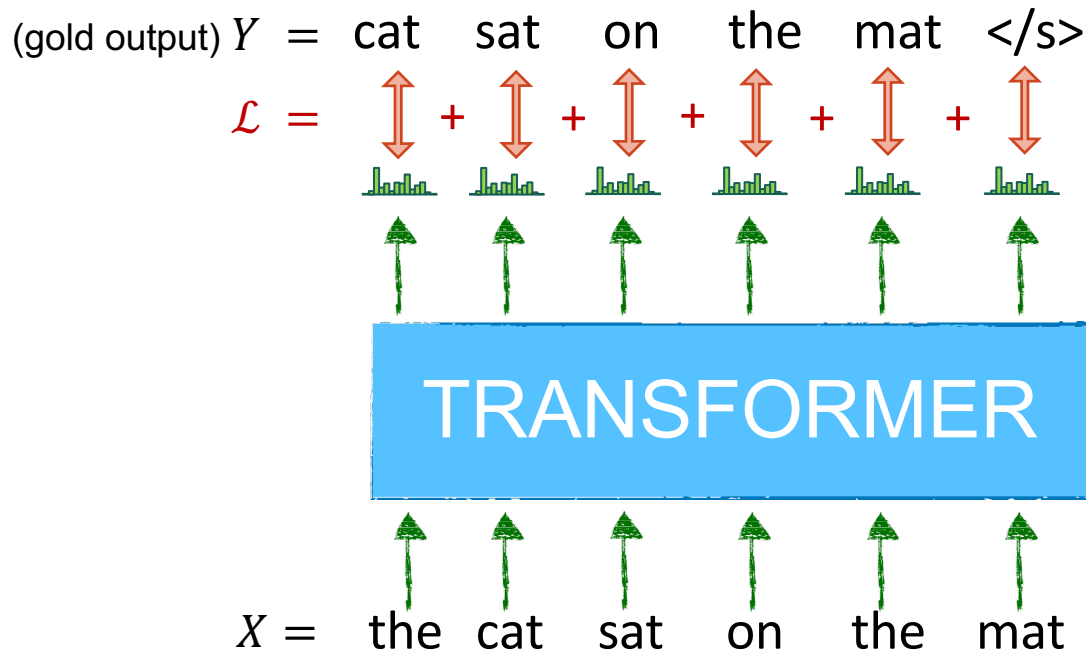
- For each position, compute the **loss** between the distribution and the gold output label.

(gold output) $Y =$ cat sat on the mat $</s>$



Training a Transformer Language Model

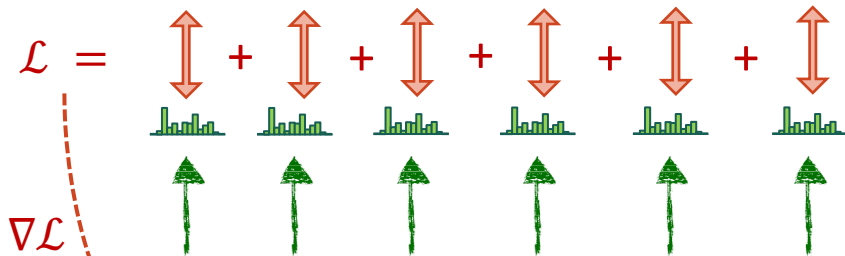
- Sum the position-wise loss values to obtain a **global loss**.



Training a Transformer Language Model

- Using this loss, do **Backprop** and **update** the Transformer parameters.

(gold output) $Y = \text{cat sat on the mat } \langle /s \rangle$



TRANSFORMER

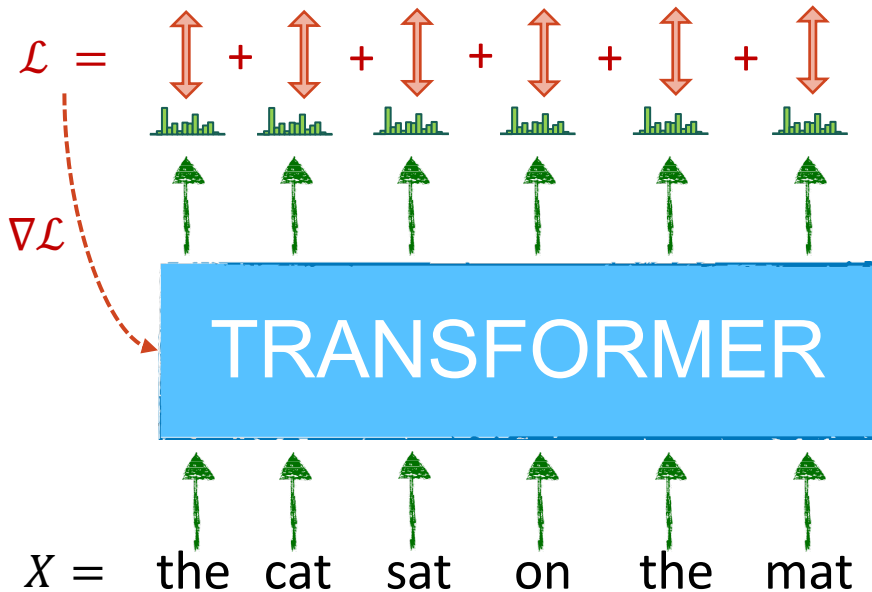
$X = \text{the cat sat on the mat}$

Well, this is not quite right 🤔 ...
what is the problem with this?

Training a Transformer Language Model

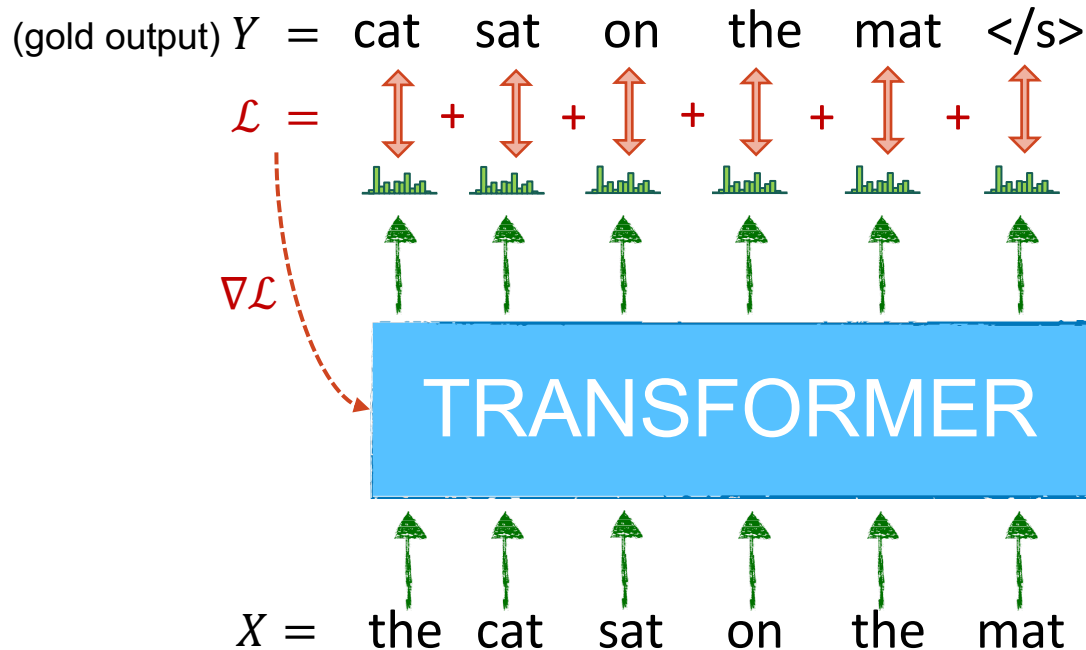
- The model would solve the task by **copying** the next token to output (data leakage).
 - Does **not** learn anything useful

(gold output) $Y =$ cat sat on the mat $</s>$



Training a Transformer Language Model

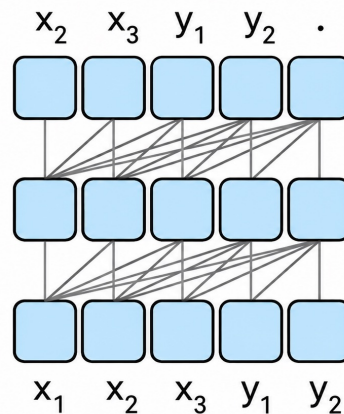
- We need to **prevent information leakage** from future tokens! How?



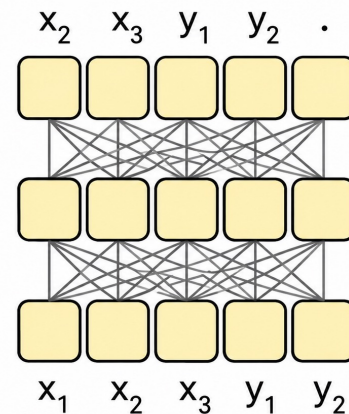
Attention mask

Attention raw scores

0	-0.08	1.24	0.69	-0.98	1.43	-0.6	0.7	0.16	0.93	1.28	-1.61	-1.1
1	-0.09	-0.0	-0.7	0.06	0.25	0.23	0.26	0.18	0.78	-0.21	-1.01	1.01
2	0.86	1.19	1.59	0.86	-0.13	-0.15	-2.13	-0.98	-0.87	-1.72	1.87	-0.72
3	0.12	-0.03	-0.02	0.88	-0.46	-0.7	0.54	-0.42	-1.89	-0.38	0.04	-0.84
4	0.51	0.17	0.13	-1.64	0.24	-0.02	1.68	-0.36	0.64	0.36	0.27	0.66
5	0.24	-1.44	0.43	0.74	0.96	-1.21	-0.31	1.54	1.66	1.14	0.58	-1.44
6	0.26	-0.1	0.93	0.72	-0.38	1.65	0.47	-0.96	-0.17	-0.9	-1.57	0.22
7	-0.55	0.81	0.71	1.7	-0.8	-1.14	-0.32	1.78	-0.7	-0.04	1.54	0.81
8	0.74	-0.76	-0.44	-0.08	-1.38	-0.13	1.25	-1.37	1.84	0.3	0.57	0.74
9	-0.97	-0.91	0.15	0.35	-0.81	0.11	1.14	-1.52	1.06	1.87	0.5	-0.3
10	1.56	0.9	0.39	1.46	1.44	-1.05	0.9	-0.73	0.36	-0.67	-0.62	-0.43
11	0.32	0.74	0.44	-0.1	1.19	0.83	0.29	2.06	0.51	-0.26	1.51	0.11
	1	2	3	4	5	6	7	8	9	10	11	12

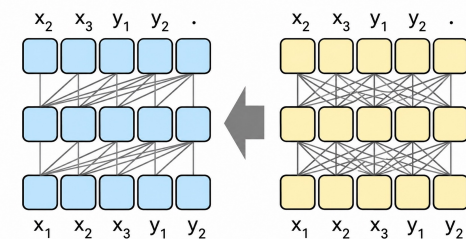


What we want



What we have

Attention mask



Attention raw scores

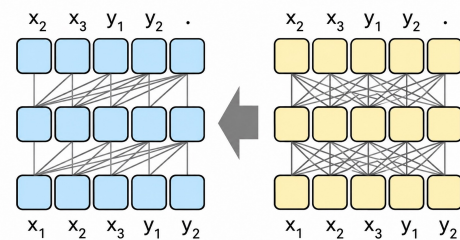
0	-0.08	1.24	0.69	-0.98	1.43	-0.6	0.7	0.16	0.93	1.28	-1.61	-1.1
1	-0.09	-0.0	-0.7	0.06	0.25	0.23	0.26	0.18	0.78	-0.21	-1.01	1.01
2	0.86	1.19	1.59	0.86	-0.13	-0.15	-2.13	-0.98	-0.87	-1.72	1.87	-0.72
3	0.12	-0.03	-0.02	0.88	-0.46	-0.7	0.54	-0.42	-1.89	-0.38	0.04	-0.84
4	0.51	0.17	0.13	-1.64	0.24	-0.02	1.68	-0.36	0.64	0.36	0.27	0.66
5	0.24	-1.44	0.43	0.74	0.96	-1.21	-0.31	1.54	1.66	1.14	0.58	-1.44
6	0.26	-0.1	0.93	0.72	-0.38	1.65	0.47	-0.96	-0.17	-0.9	-1.57	0.22
7	-0.55	0.81	0.71	1.7	-0.8	-1.14	-0.32	1.78	-0.7	-0.04	1.54	0.81
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9	-0.97	-0.91	0.15	0.35	-0.81	0.11	1.14	-1.52	1.06	1.87	0.5	-0.3
10	1.56	0.9	0.39	1.46	1.44	-1.05	0.9	-0.73	0.36	-0.67	-0.62	-0.43
11	0.32	0.74	0.44	-0.1	1.19	0.83	0.29	2.06	0.51	-0.26	1.51	0.11
	1	2	3	4	5	6	7	8	9	10	11	12

Attention mask

0	0.0	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf
1	0.0	0.0	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf
2	0.0	0.0	0.0	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf
3	0.0	0.0	0.0	0.0	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf
4	0.0	0.0	0.0	0.0	0.0	-inf	-inf	-inf	-inf	-inf	-inf	-inf
5	0.0	0.0	0.0	0.0	0.0	0.0	-inf	-inf	-inf	-inf	-inf	-inf
6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-inf	-inf	-inf	-inf	-inf
7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-inf	-inf	-inf	-inf
8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-inf	-inf	-inf
9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-inf	-inf
10	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-inf
11	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	0	1	2	3	4	5	6	7	8	9	10	11

large negative numbers,
which leads to $\text{softmax}(-\infty) \approx 0$

Attention mask



Attention raw scores

0	-0.08	1.24	0.69	-0.98	1.43	-0.6	0.7	0.16	0.93	1.28	-1.61	-1.1
1	-0.09	-0.0	-0.7	0.06	0.25	0.23	0.26	0.18	0.78	-0.21	-1.01	1.01
2	0.86	1.19	1.59	0.86	-0.13	-0.15	-2.13	-0.98	-0.87	-1.72	1.87	-0.72
3	0.12	-0.03	-0.02	0.88	-0.46	-0.7	0.54	-0.42	-1.89	-0.38	0.04	-0.84
4	0.51	0.17	0.13	-1.64	0.24	-0.02	1.68	-0.36	0.64	0.36	0.27	0.66
5	0.24	-1.44	0.43	0.74	0.96	-1.21	-0.31	1.54	1.66	1.14	0.58	-1.44
6	0.26	-0.1	0.93	0.72	-0.38	1.65	0.47	-0.96	-0.17	-0.9	-1.57	0.22
7	-0.55	0.81	0.71	1.7	-0.8	-1.14	-0.32	1.78	-0.7	-0.04	1.54	0.81
8	0.74	-0.76	-0.44	-0.08	-1.38	-0.13	1.25	-1.37	1.84	0.3	0.57	0.74
9	-0.97	-0.91	0.15	0.35	-0.81	0.11	1.14	-1.52	1.06	1.87	0.5	-0.3
10	1.56	0.9	0.39	1.46	1.44	-1.05	0.9	-0.73	0.36	-0.67	-0.62	-0.43
11	0.32	0.74	0.44	-0.1	1.19	0.83	0.29	2.06	0.51	-0.26	1.51	0.11
	1	2	3	4	5	6	7	8	9	10	11	12

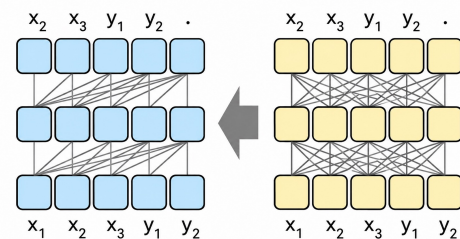
+

Apply
the mask
element-wise

Attention mask

0	0.0	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf
1	0.0	0.0	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf
2	0.0	0.0	0.0	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf
3	0.0	0.0	0.0	0.0	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf
4	0.0	0.0	0.0	0.0	0.0	-inf	-inf	-inf	-inf	-inf	-inf	-inf
5	0.0	0.0	0.0	0.0	0.0	0.0	-inf	-inf	-inf	-inf	-inf	-inf
6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-inf	-inf	-inf	-inf	-inf
7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-inf	-inf	-inf	-inf
8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-inf	-inf	-inf
9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-inf	-inf
10	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-inf
11	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	0	1	2	3	4	5	6	7	8	9	10	11

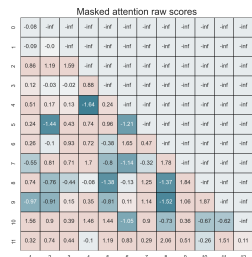
Attention mask



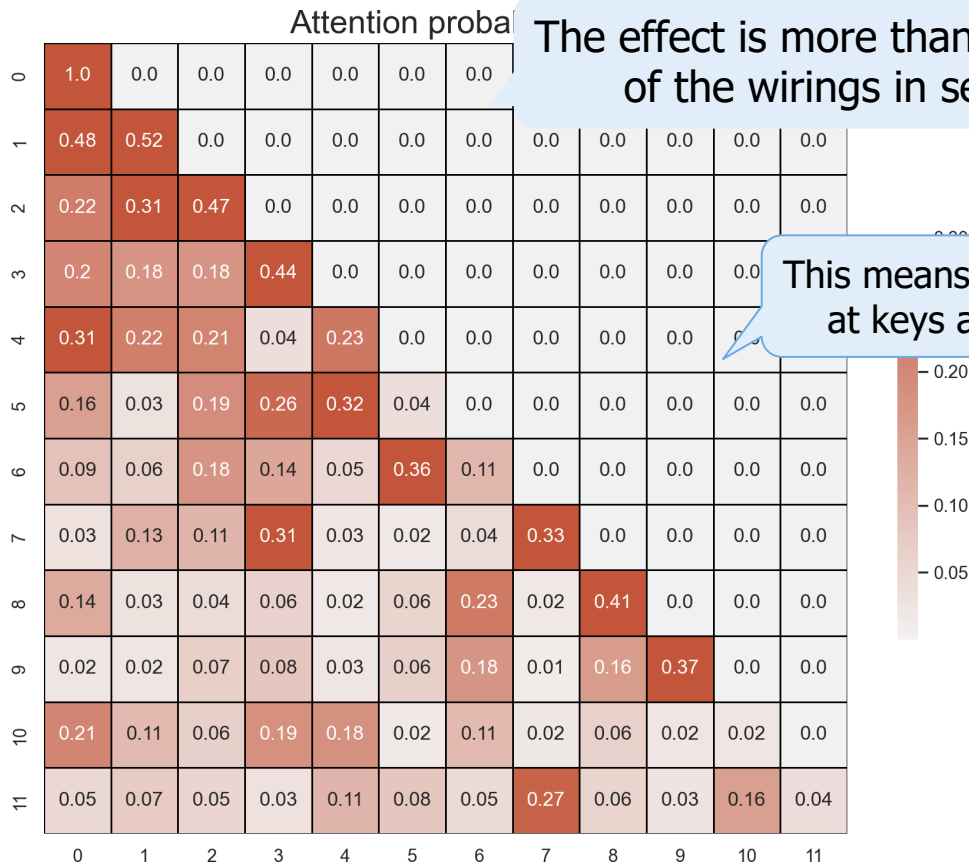
Masked attention raw scores

0	-0.08	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf
1	-0.09	-0.0	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf
2	0.86	1.19	1.59	-inf	-inf	-inf	-inf	-inf	-inf	-inf	-inf
3	0.12	-0.03	-0.02	0.88	-inf	-inf	-inf	-inf	-inf	-inf	-inf
4	0.51	0.17	0.13	-1.64	0.24	-inf	-inf	-inf	-inf	-inf	-inf
5	0.24	-1.44	0.43	0.74	0.96	-1.21	-inf	-inf	-inf	-inf	-inf
6	0.26	-0.1	0.93	0.72	-0.38	1.65	0.47	-inf	-inf	-inf	-inf
7	-0.55	0.81	0.71	1.7	-0.8	-1.14	-0.32	1.78	-inf	-inf	-inf
8	0.74	-0.76	-0.44	-0.08	-1.38	-0.13	1.25	-1.37	1.84	-inf	-inf
9	-0.97	-0.91	0.15	0.35	-0.81	0.11	1.14	-1.52	1.06	1.87	-inf
10	1.56	0.9	0.39	1.46	1.44	-1.05	0.9	-0.73	0.36	-0.67	-0.62
11	0.32	0.74	0.44	-0.1	1.19	0.83	0.29	2.06	0.51	-0.26	1.51
	1	2	3	4	5	6	7	8	9	10	11

Attention mask



softmax



The effect is more than just pruning out some of the wirings in self-attention block.

This means that the query-3 can look at keys at positions {0, 1, 2, 3}.

Attention mask: a real example

```
scores = (q @ k.transpose(-2, -1)) / math.sqrt(q.size(-1))

# Mask entries above the diagonal.
mask = torch.triu(
    torch.ones_like(scores, dtype=torch.bool), diagonal=1
)
scores = scores.masked_fill(mask, float("-inf"))

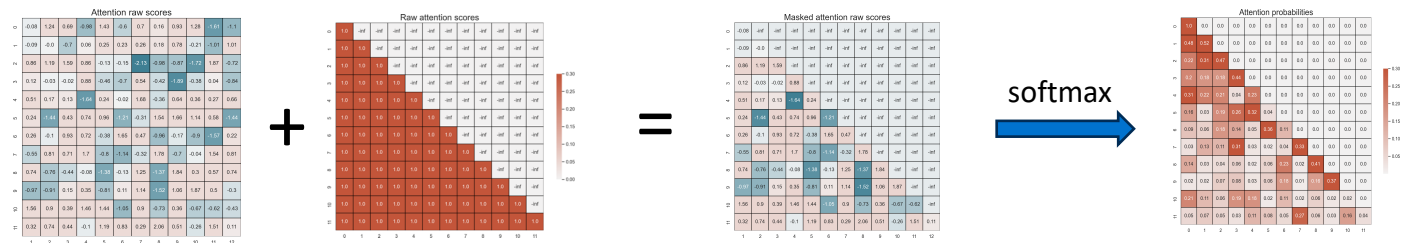
weights = torch.softmax(scores, dim=-1)
output = weights @ v
```

(2) The construction of the mask tensor is a bit different than our slides in practice.

(1) We're using a masking-specific function, not tensor addition. Why?

Attention masking: Why Before Softmax?

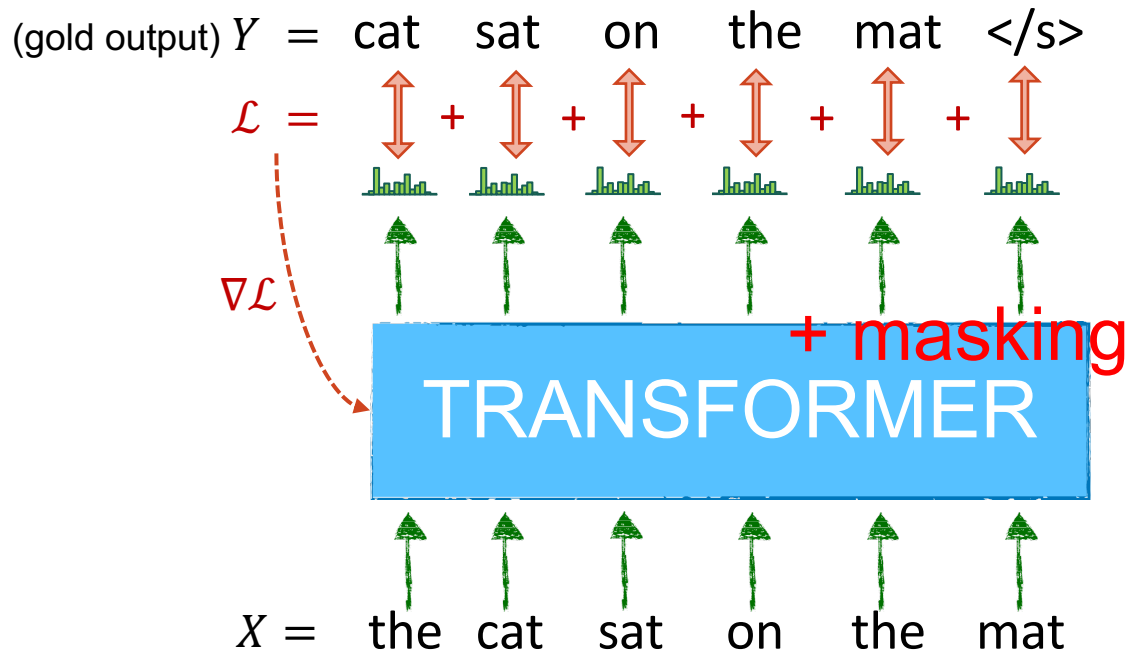
- We applied attention masking before softmax. Why not after?



- Softmax normalizes the scores so that it's a probability.
- Masking after softmax, would lead to an unnormalized probability distribution. So, you have to normalize twice! (wasted computation which we don't want)

Training a Transformer Language Model

- We need to **prevent information leakage** from future tokens! How?



Quiz: attention masking

- What's the point of attention masking during training?
 - A. speeds up training by reducing the number of computations performed.
 - B. prevents accessing future tokens, ensuring that predictions are based only on past and present information.
 - C. selectively highlights important tokens in the input, thereby enhancing attention mechanism.

Summary: A Decoder-only Transformer

- This is a very generic Transformer!
- **Next:** A more extended version of the Transformer architecture.

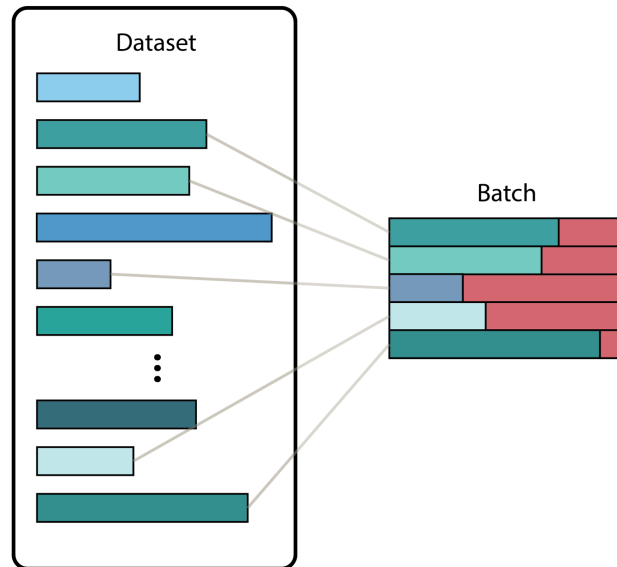
Padding sequences

- **Why pad?**

- GPUs process data in batches, i.e., tensor of shape (batch_size, seq_len).
- Sequences in a batch have different lengths;
- Padding makes their tensor shapes match.

- Padding is (often) done differently for training vs. inf.

- The two phases need different parts of the sequence.



Padding sequences

- **Inference uses left-padding:**

- Every prompt now ends in the same column
- The next token for the whole batch is read from the last position.
- We want to avoid scenario where various inputs (in the batch) need to be decoded from different dimensions.

- **Training uses right-padding:**

- Right padding is simple:
 - real content they all start at position 0 and labels line up with inputs.
- In theory, left-padding during training would also work
 - The model predicts every next token in parallel.
 - It doesn't matter where each sequence ends.

Inference · left padding

prompt (left-padded)

generated

input tokens

seq 0	The	cat	sat	on	the	mat
seq 1	[PAD]	[PAD]	I	love	deep	learning
seq 2	[PAD]	[PAD]	[PAD]	Hello	world	!
seq 3	[PAD]	Attention	is	all	you	need
	0	1	2	3	4	5
	column					

Training · right padding

input tokens

seq 0	The	cat	sat	on	the	mat	.
seq 1	I	love	deep	learning	[PAD]	[PAD]	[PAD]
seq 2	Attention	is	all	you	need	[PAD]	[PAD]
seq 3	Hello	world	[PAD]	[PAD]	[PAD]	[PAD]	[PAD]
	0	1	2	3	4	5	6
	column						

Padding sequences

- We also need to mask out padding tokens so we're not training for those PAD positions.
- Here `attention_mask` marks which tokens are real vs PAD.
- This is then applied to attention matrix:
 - The keys of PAD tokens are masked.
 - The pad queries can still attend to the real tokens (a design choice), but their outputs are thrown away since nothing requires pad queries to attend to anything.

Training · right padding

	input_ids						
seq 0	The	cat	sat	on	the	mat	.
seq 1	I	love	deep	learning	[PAD]	[PAD]	[PAD]
seq 2	Attention	is	all	you	need	[PAD]	[PAD]
seq 3	Hello	world	[PAD]	[PAD]	[PAD]	[PAD]	[PAD]
	0	1	2	3	4	5	6
	column						

	attention_mask (B, T)						
	1	1	1	1	1	1	1
	1	1	1	1	0	0	0
	1	1	1	1	0	0	0
	1	1	0	0	0	0	0
	0	1	2	3	4	5	6

seq 1: score mask (T, T)

	1	1	1	1	0	0	0
	↓	↓	↓	↓	↓	↓	↓
I	0.0	-inf	-inf	-inf	-inf	-inf	-inf
love	0.0	0.0	-inf	-inf	-inf	-inf	-inf
deep	0.0	0.0	0.0	-inf	-inf	-inf	-inf
learning	0.0	0.0	0.0	0.0	-inf	-inf	-inf
[PAD]	0.0	0.0	0.0	0.0	-inf	-inf	-inf
[PAD]	0.0	0.0	0.0	0.0	-inf	-inf	-inf
[PAD]	0.0	0.0	0.0	0.0	-inf	-inf	-inf
I	0.0	-inf	-inf	-inf	-inf	-inf	-inf
love	0.0	0.0	-inf	-inf	-inf	-inf	-inf
deep	0.0	0.0	0.0	-inf	-inf	-inf	-inf
learning	0.0	0.0	0.0	0.0	-inf	-inf	-inf
[PAD]	0.0	0.0	0.0	0.0	-inf	-inf	-inf
[PAD]	0.0	0.0	0.0	0.0	-inf	-inf	-inf
[PAD]	0.0	0.0	0.0	0.0	-inf	-inf	-inf

query

Padding sequences

- The logic is the same for inference: the keys of PAD tokens are masked since these tokens are uninformative.

Inference • left padding

prompt (left-padded)

generated

input_ids

	0	1	2	3	4	5
seq 0	The	cat	sat	on	the	mat
seq 1	[PAD]	[PAD]	I	love	deep	learning
seq 2	[PAD]	[PAD]	[PAD]	Hello	world	!
seq 3	[PAD]	Attention	is	all	you	need

attention_mask (B, T)

	0	1	2	3	4	5
seq 0	1	1	1	1	1	1
seq 1	0	0	1	1	1	1
seq 2	0	0	0	1	1	1
seq 3	0	1	1	1	1	1

seq 1: score mask (T, T)

	0	1	2	3	4	5
[PAD]	-inf	-inf	-inf	-inf	-inf	-inf
[PAD]	-inf	-inf	-inf	-inf	-inf	-inf
I	-inf	-inf	0.0	-inf	-inf	-inf
love	-inf	-inf	0.0	0.0	-inf	-inf
deep	-inf	-inf	0.0	0.0	0.0	-inf
learning	-inf	-inf	0.0	0.0	0.0	0.0

Demo time!!

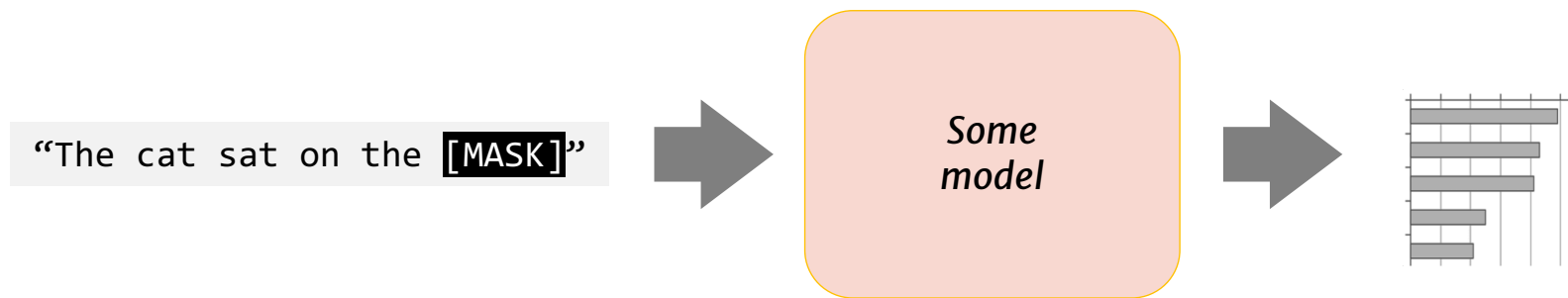
- <https://poloclub.github.io/transformer-explainer/>



Transformer Architectural Variants

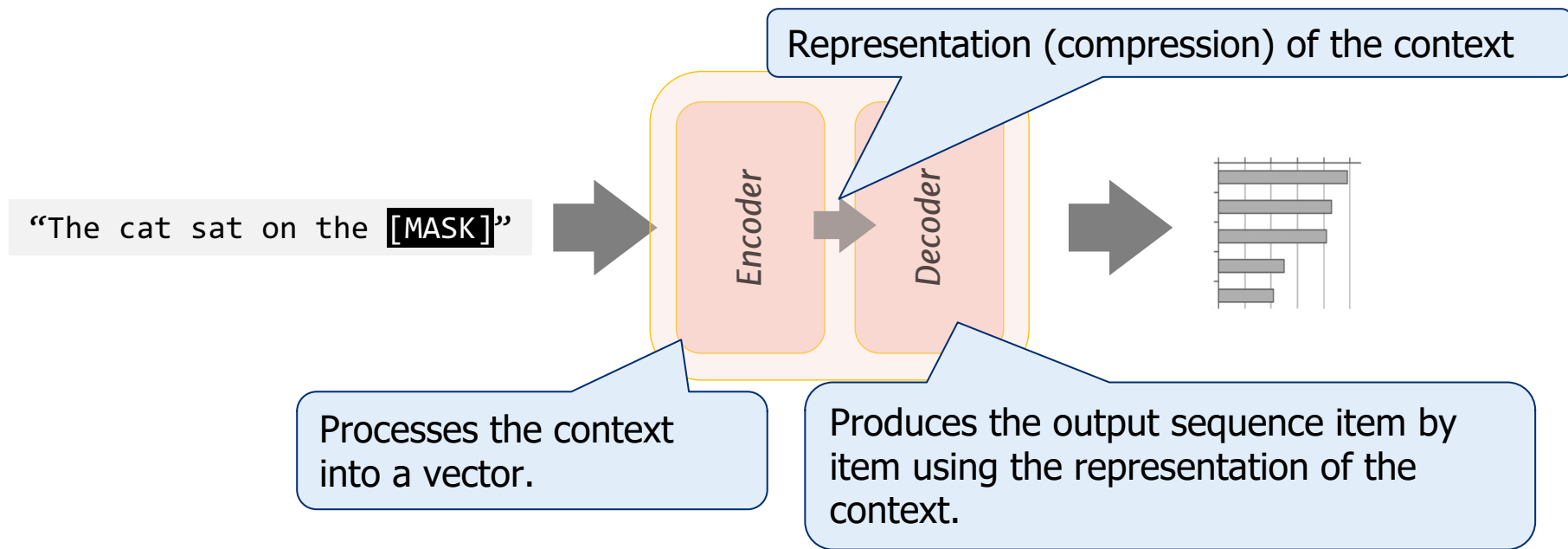
Encoder-Decoder Architectures

- It is useful to think of generative models as two sub-models.



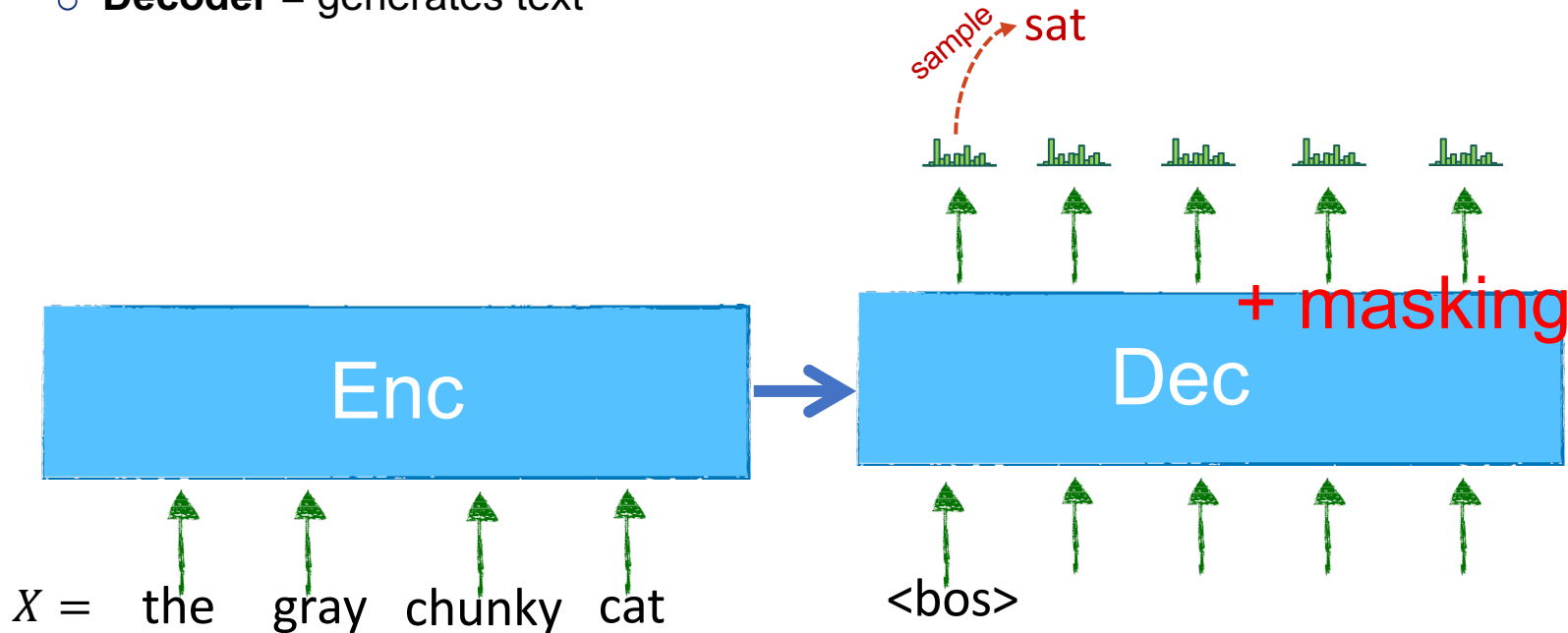
Encoder-Decoder Architectures

- It is useful to think of generative models as two sub-models.



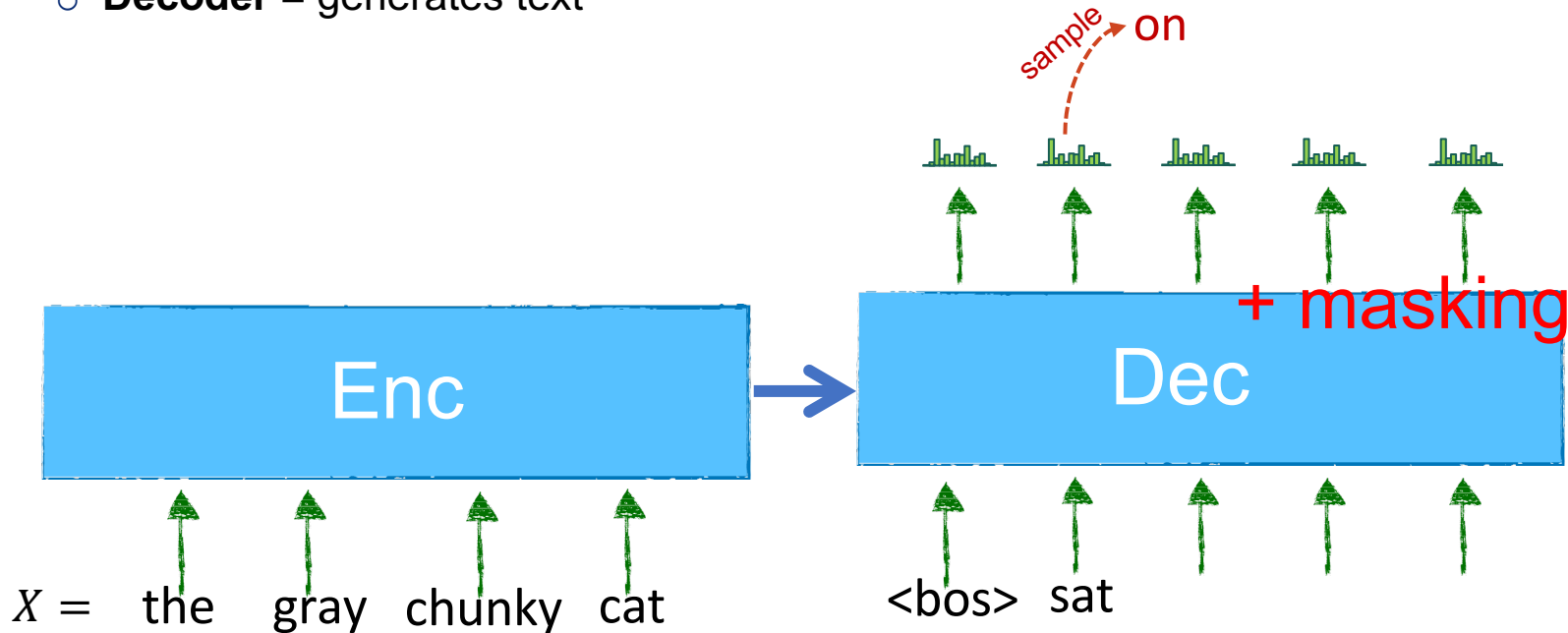
Enc-Dec work at inference time

- Transformer is two blocks
 - Encoder** = read or encode the input
 - Decoder** = generates text



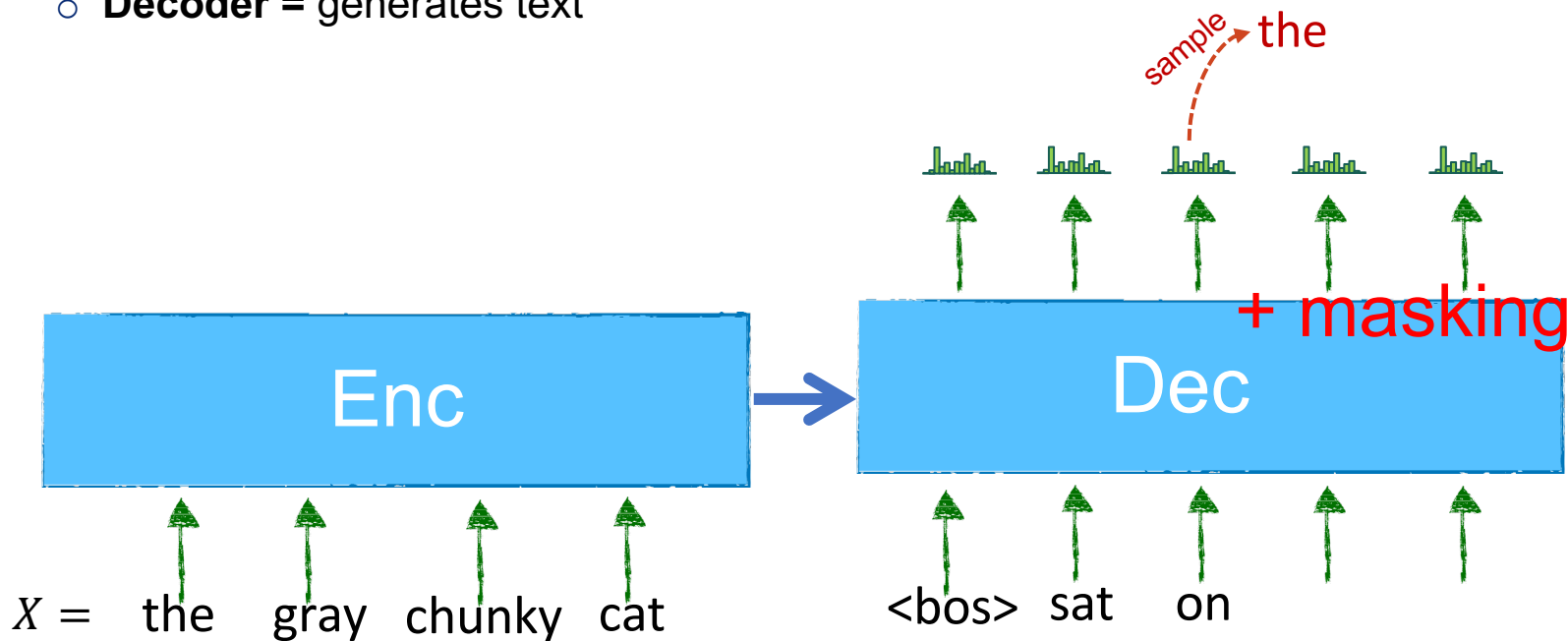
Enc-Dec work at inference time

- Transformer is two blocks
 - Encoder** = read or encode the input
 - Decoder** = generates text



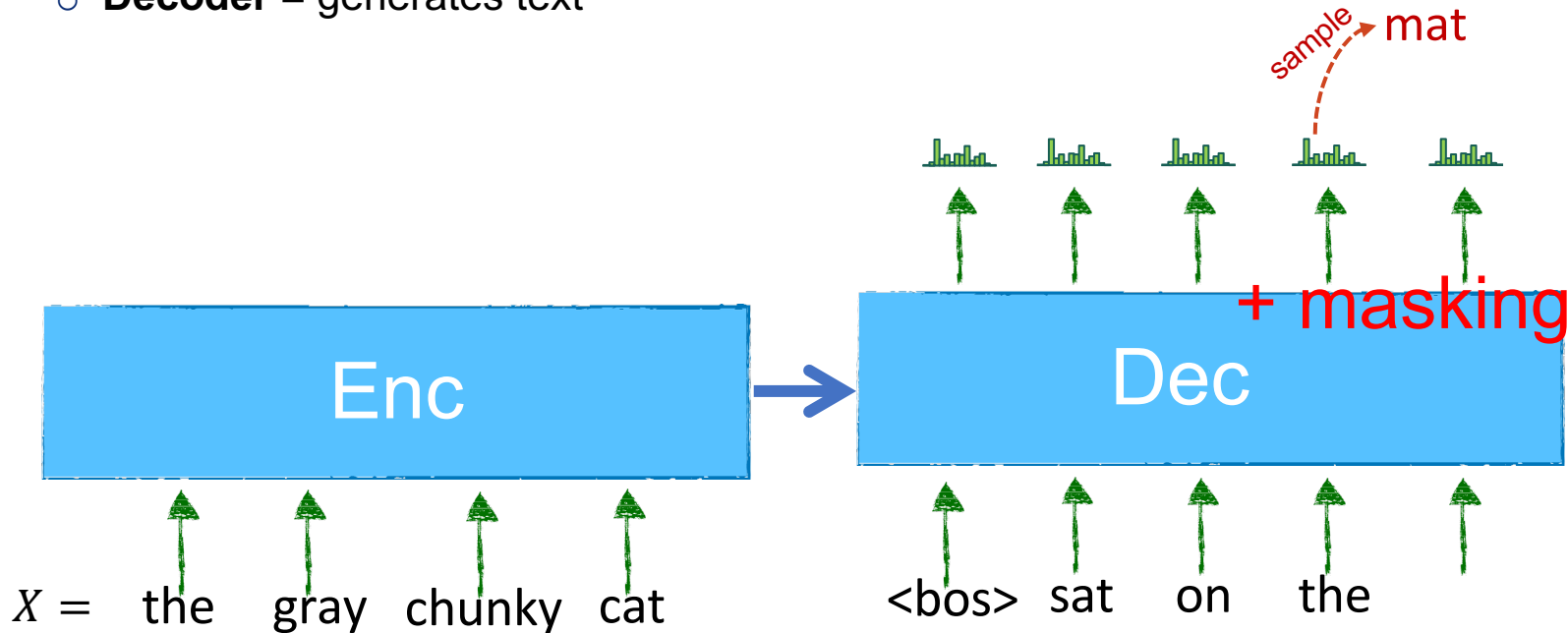
Enc-Dec work at inference time

- Transformer is two blocks
 - Encoder** = read or encode the input
 - Decoder** = generates text



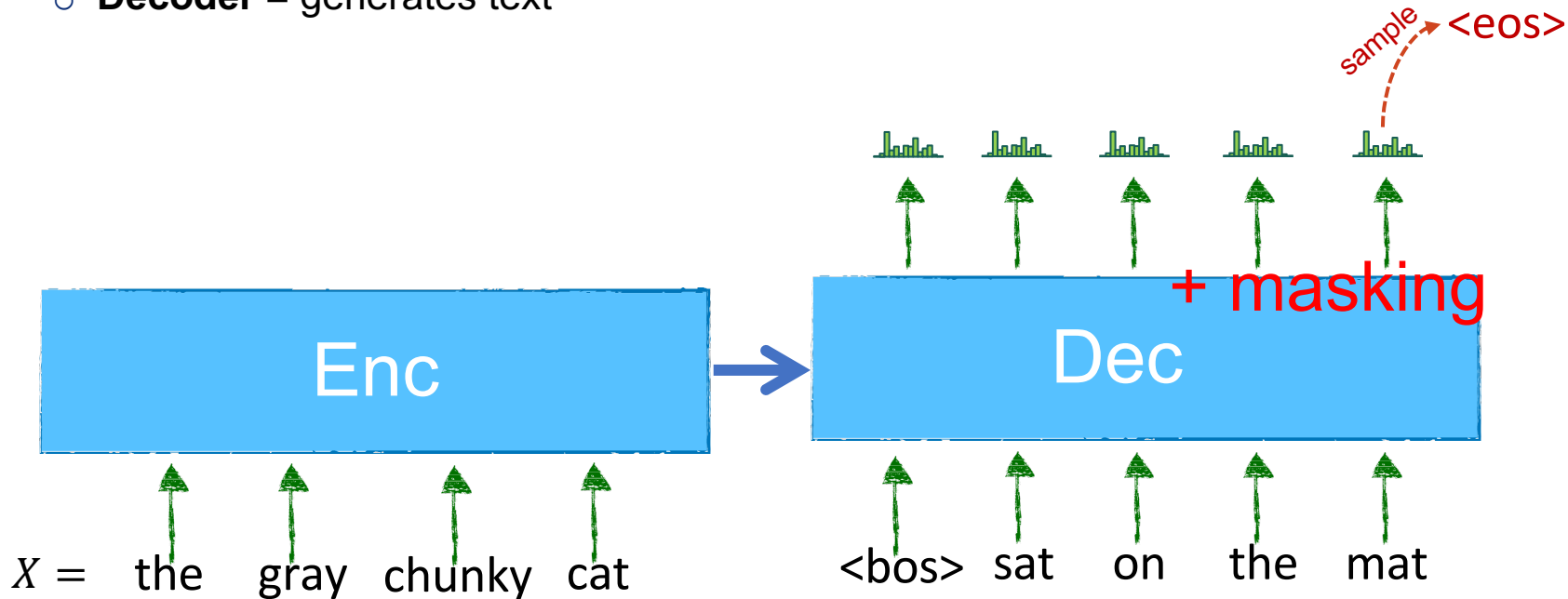
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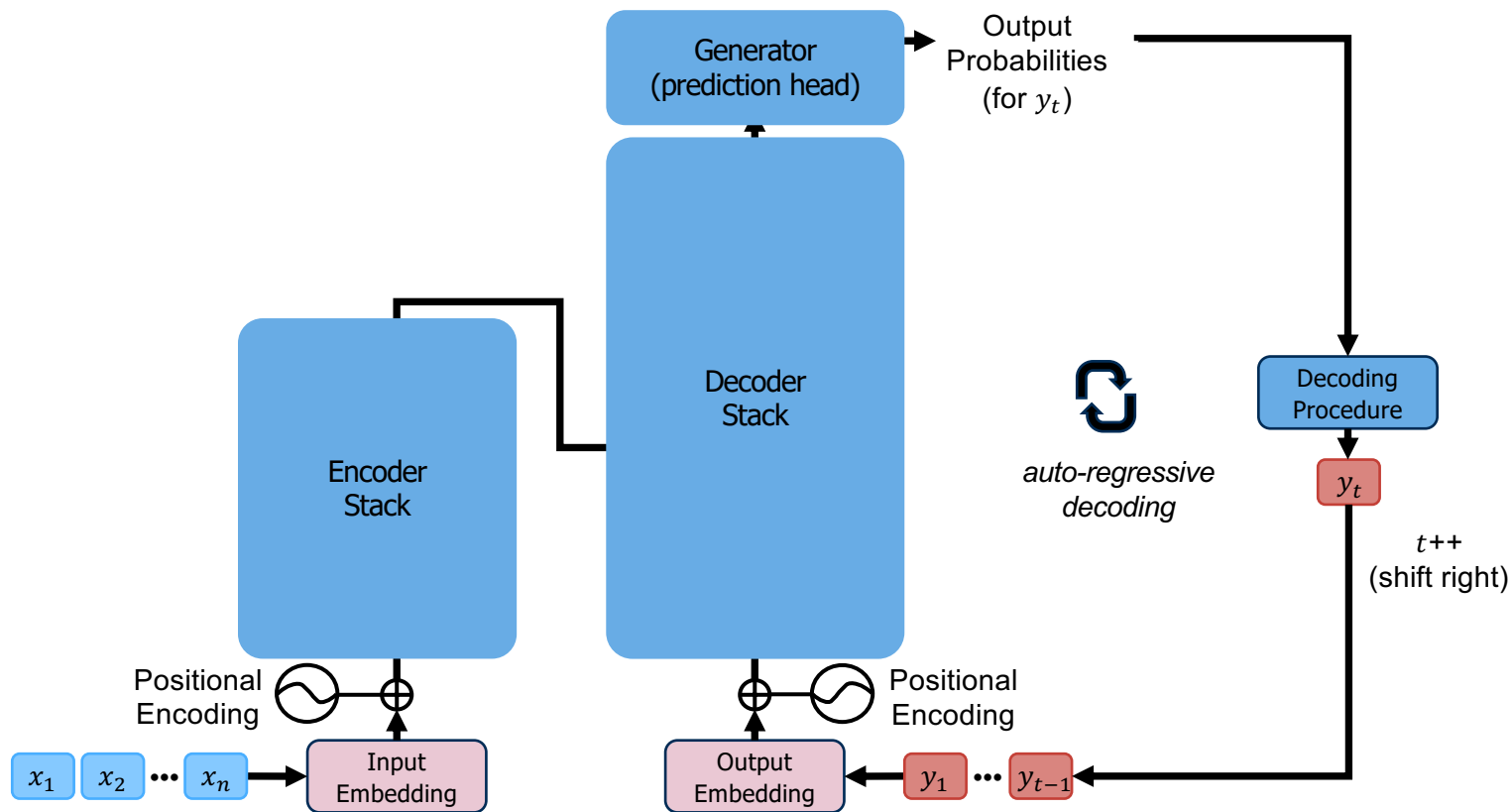


Enc-Dec work at inference time

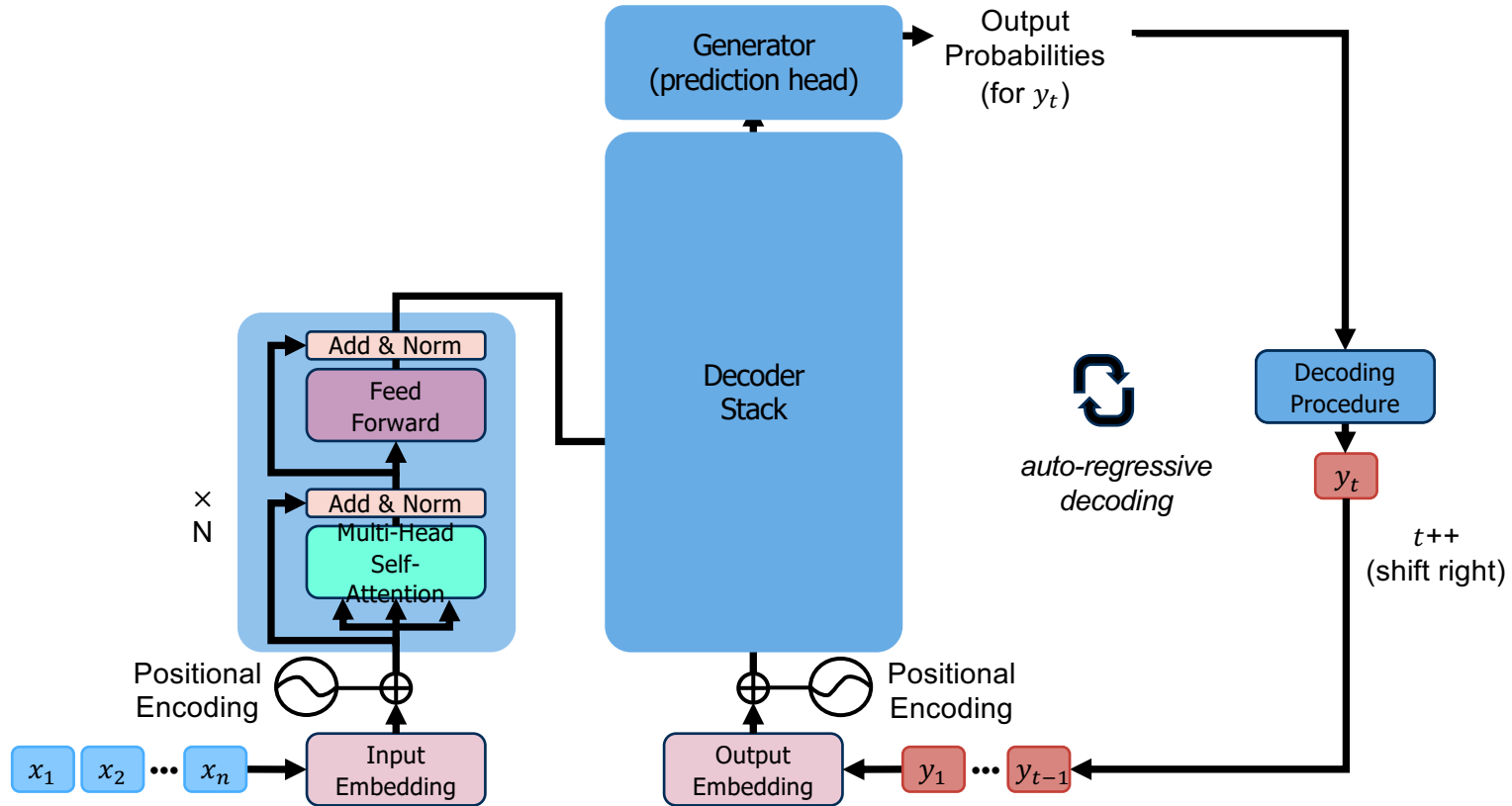
- Transformer is two blocks
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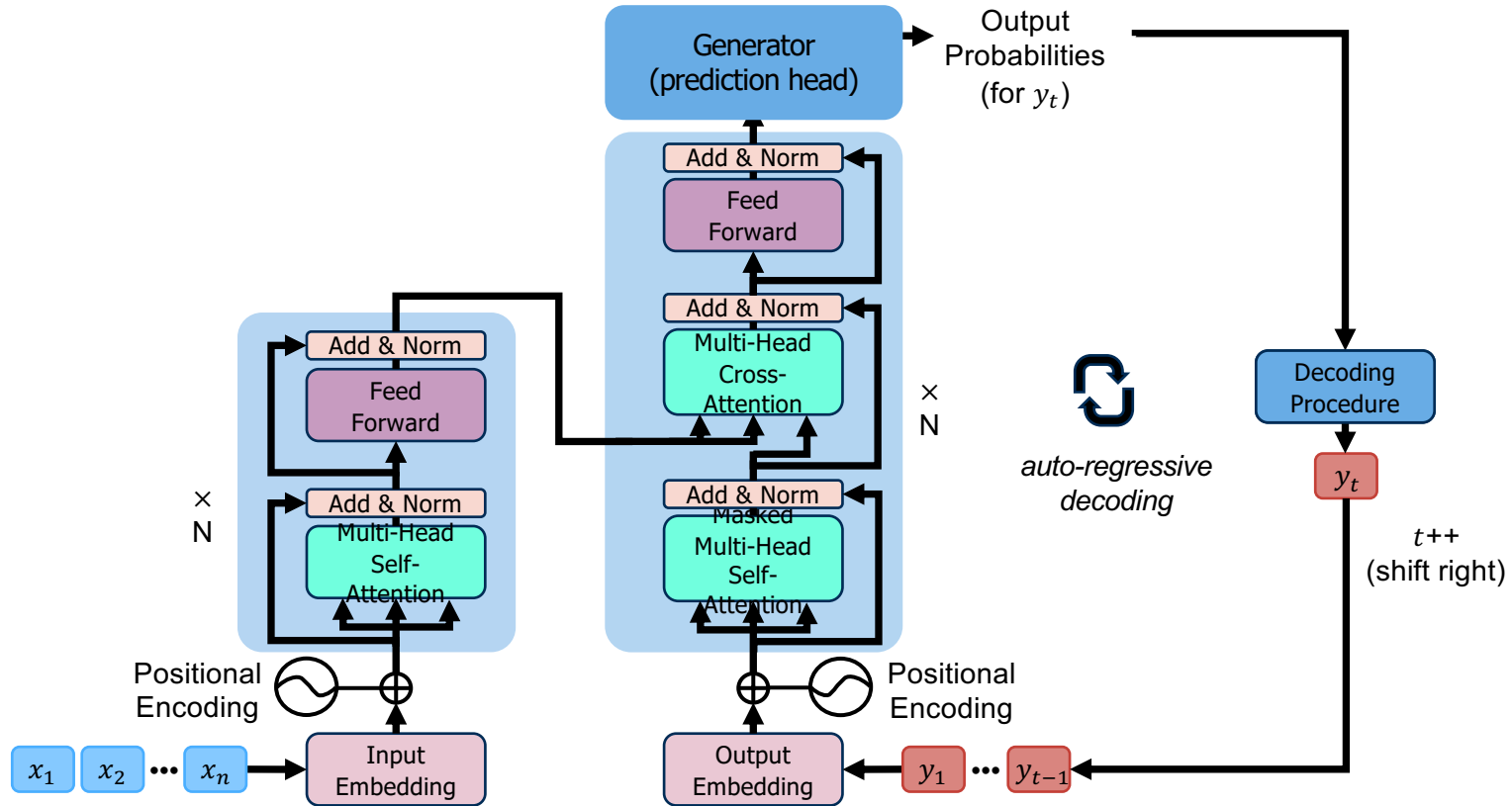
Transformer [Vaswani et al. 2017]



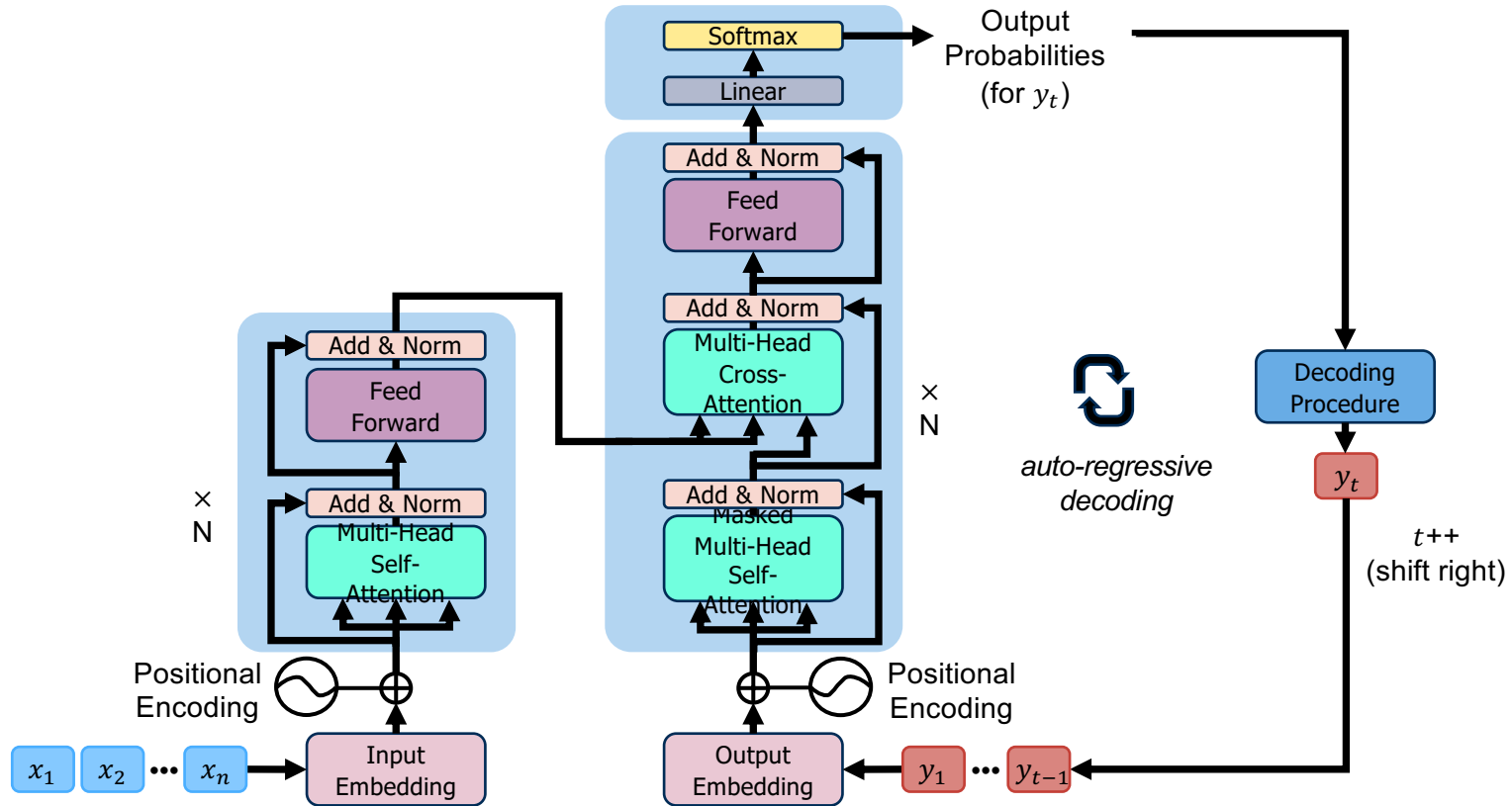
Transformer [Vaswani et al. 2017]



Transformer [Vaswani et al. 2017]

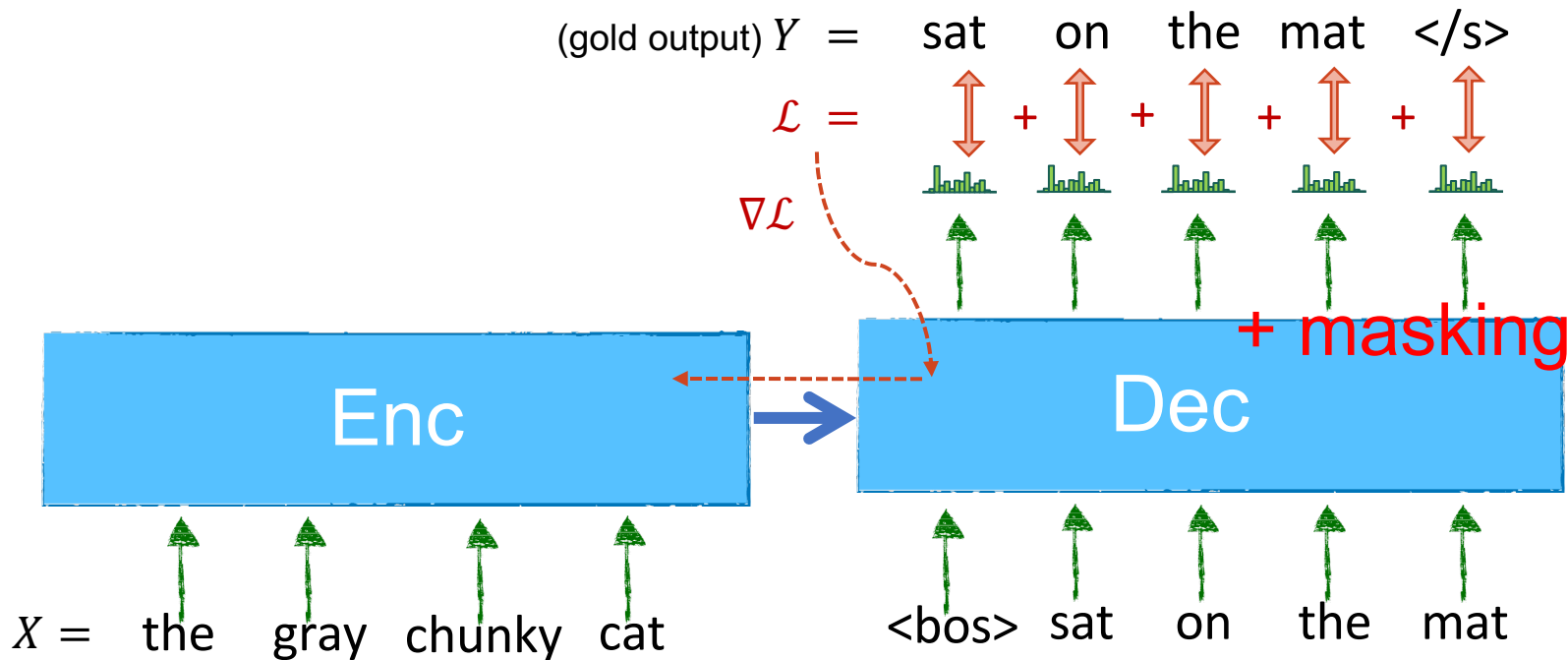


Transformer [Vaswani et al. 2017]



How does training Enc-Dec work?

- Same as what we did before for encoder-only models.

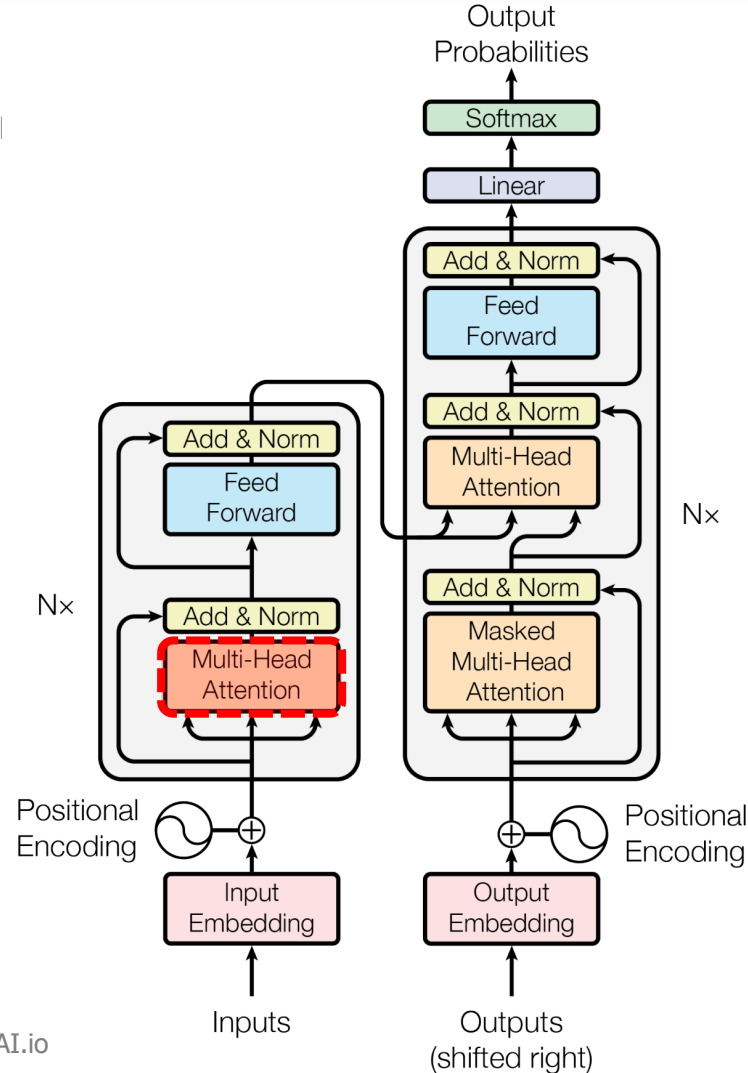
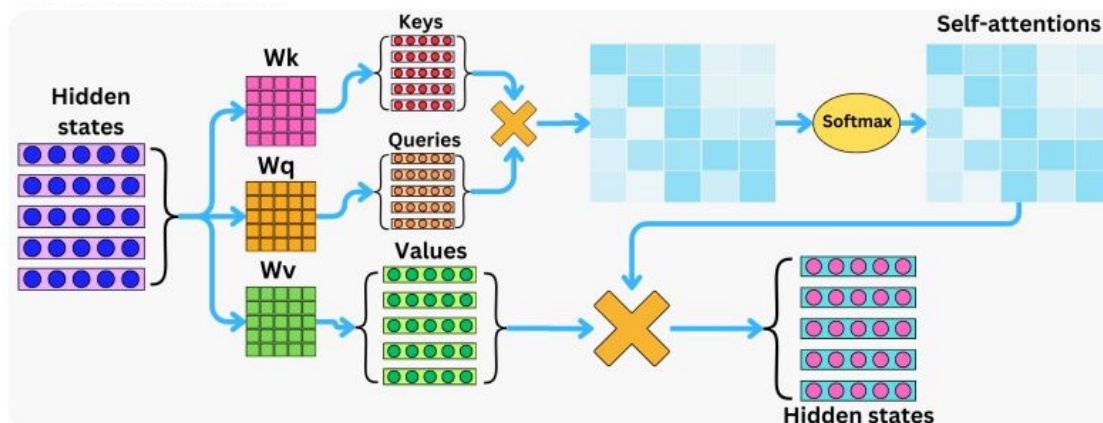


Transformer [Vaswani

- Computation of **encoder** attends to both sides.

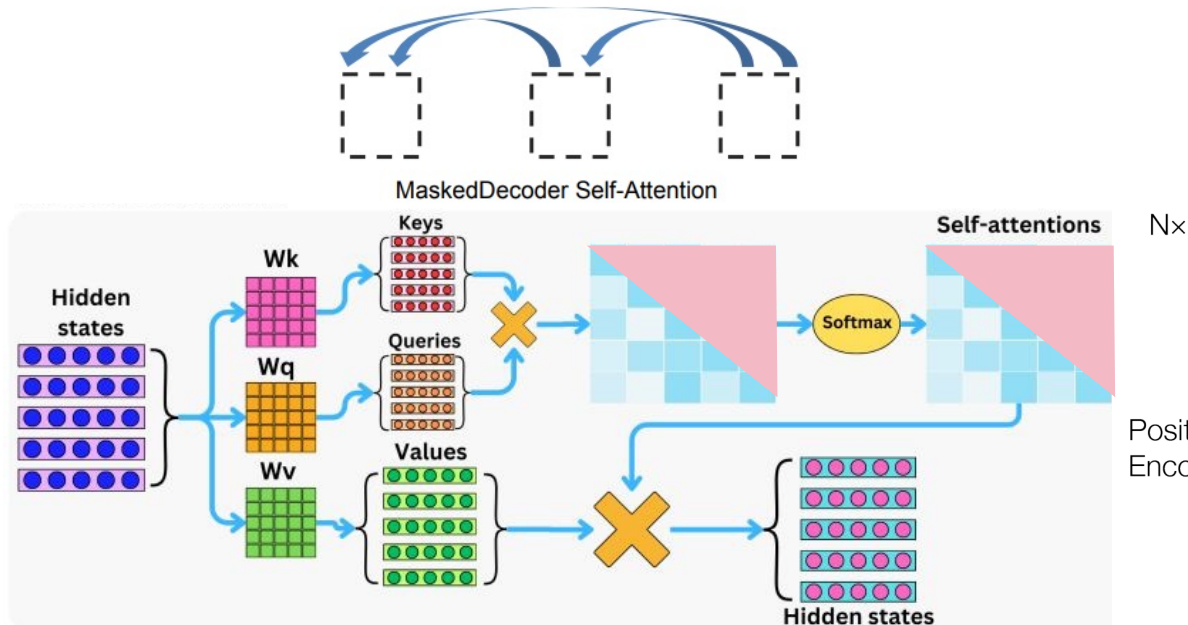


Encoder Self-Attention



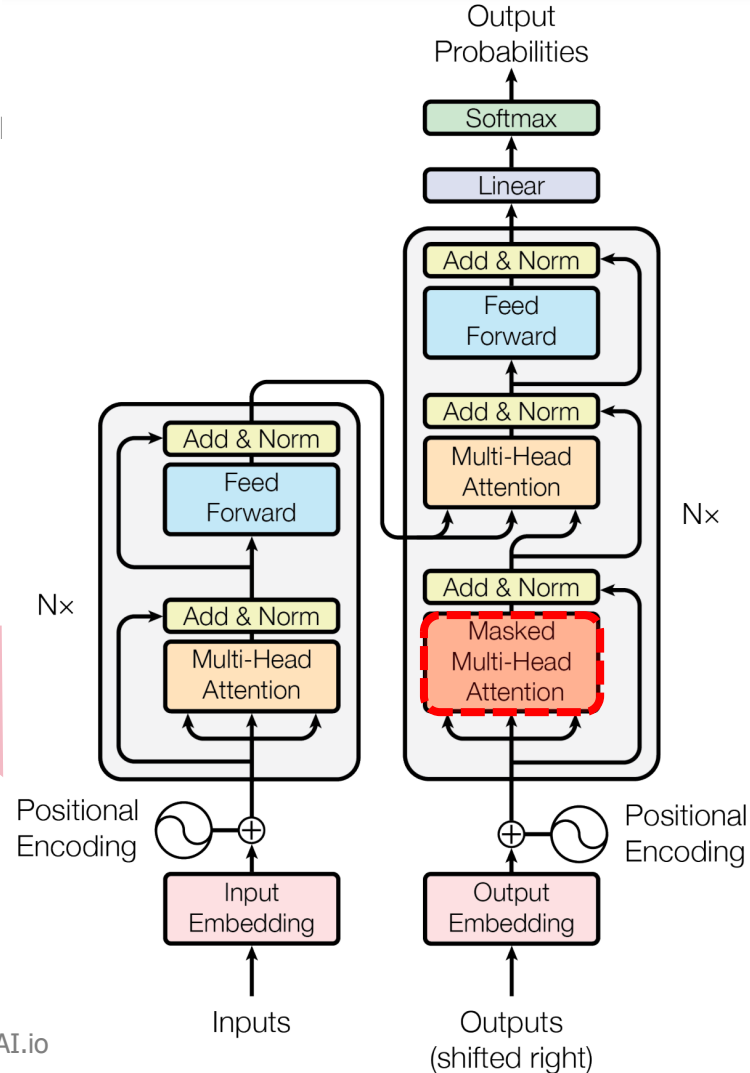
Transformer [Vaswani

- At any step of **decoder**, it attends to previous computation of **encoder** as well as **decoder's** own generations.



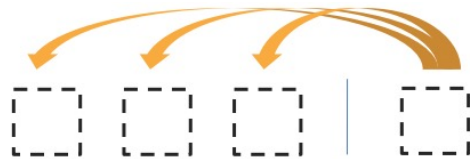
[[Attention Is All You Need, Vaswani et al. 2017](#)]

Visualization: theEdgeAI.io

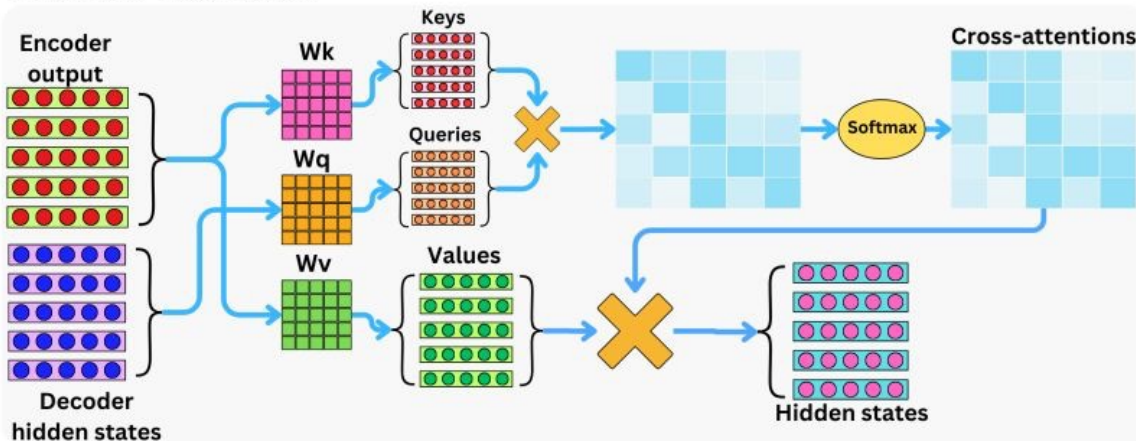


Transformer [Vaswani

- At any step of **decoder**, it attends to previous computation of **encoder**.

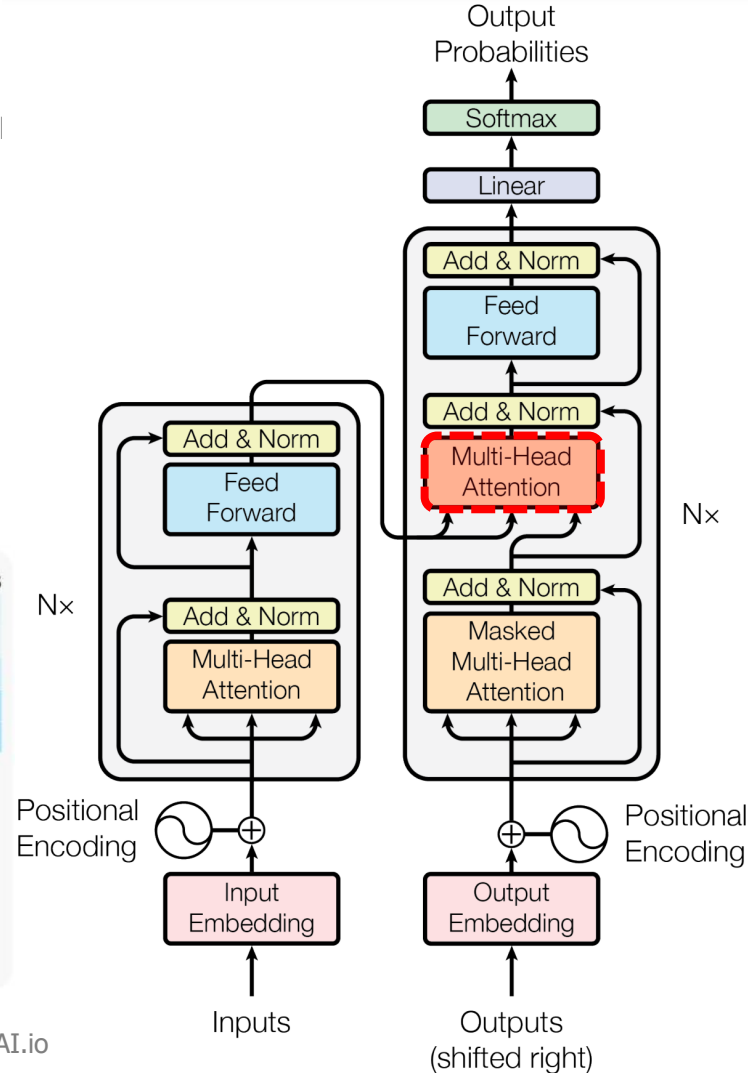


Encoder-Decoder Attention



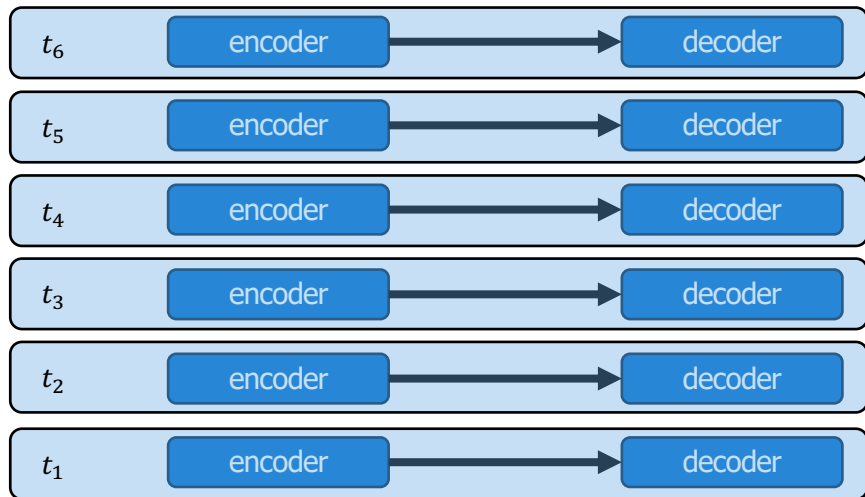
[[Attention Is All You Need, Vaswani et al. 2017](#)]

Visualization: theEdgeAI.io

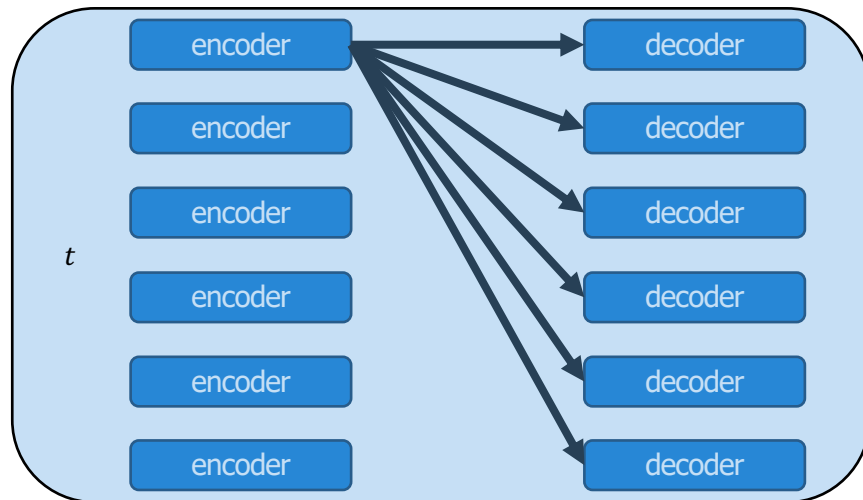


Quiz: Enc-Dec Connections

- Which best represents encoder-decoder connections? (y-axis denotes the layers)



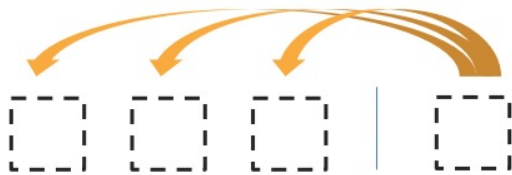
Incorrect



Correct

Recap: Transformer

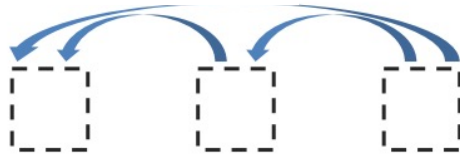
- Yaaay we know Transformers now! 🥳
- An **encoder-decoder** architecture
- 3 forms of attention



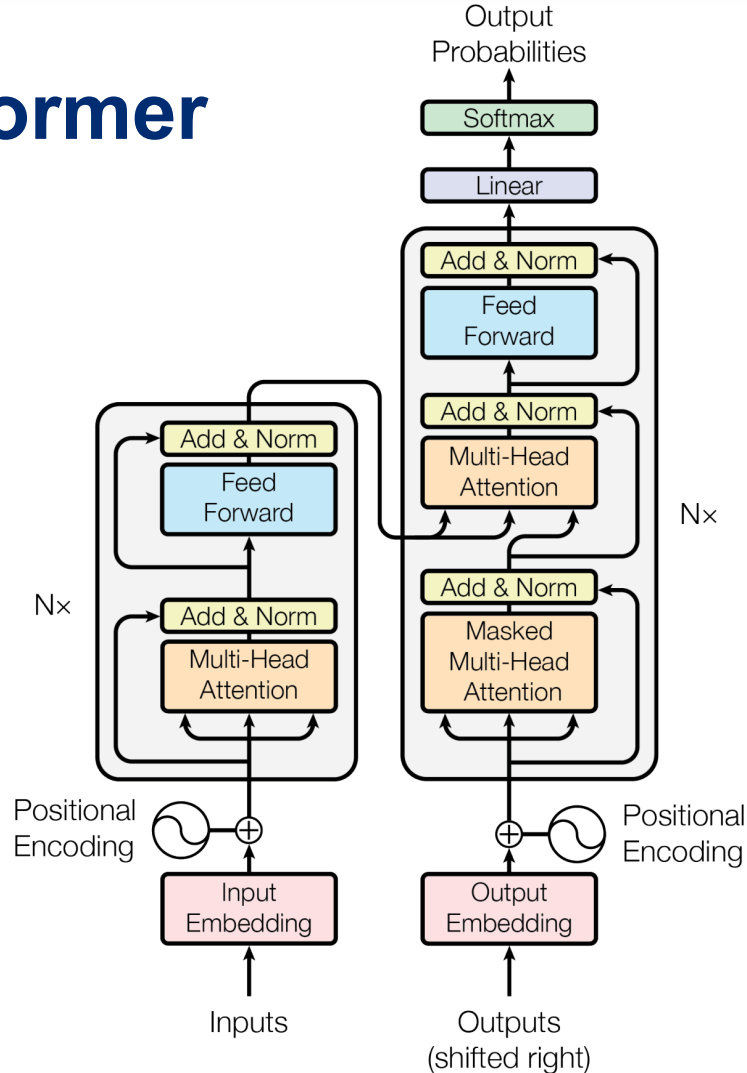
Encoder-Decoder Attention



Encoder Self-Attention



Masked Decoder Self-Attention



Considerations about computational cost in Transformers

Diversion: Floating-point Ops: FLOPS

- Floating point operations per second (FLOPS, flops or flop/s)
- Each FLOP can represent an addition, subtraction, multiplication, or division of floating-point numbers,
- The total FLOP of a model (e.g., Transformer) provides a basic approximation of computational costs associated with that model.

FLOPS of Matrix Multiplication

- Matrix-vector multiplication are common in Self-Attention (e.g., QKV projection)
 - Requires $2mn$ (2 x matrix size) operations for multiplying $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^n$
 - (2 because 1 for multiplication, 1 for addition)

$$\begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \cdots & A_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} A_{11}x_1 + A_{12}x_2 + \cdots + A_{1n}x_n \\ A_{21}x_1 + A_{22}x_2 + \cdots + A_{2n}x_n \\ \vdots \\ A_{m1}x_1 + A_{m2}x_2 + \cdots + A_{mn}x_n \end{bmatrix}$$

Quiz: thinking about computations

- Consider the following matrix multiplication:
 AB , where $A \in \mathbb{R}^{n \times m}, B \in \mathbb{R}^{m \times k}$
- **Question 1:** Computing AB involves how many **arithmetic operations**?
 - (also referred to as floating-point operations or FLOPs)
- **Answer:** It's $O(n \times m \times k)$.
 - (to be a bit more precise, it's $\approx 2n \times m \times k$ since each element in AB requires almost equal num of multiplications and summations.)
- **Question 2:** Computing AB involves how many memory/IO access?
- **Answer:** It's $n \times m$ (reading A) + $m \times k$ (reading B) + $n \times k$ (writing AB).

Computations in Self-Attention Block

- We are going to count computations and IO access in Transformer computations.
- Note we assume that the full input sequence is given at once (e.g., training time).
- Here is the first step. Given: $\mathbf{x} \in \mathbb{R}^{b \times n \times d}$, $\mathbf{W}_i^q \in \mathbb{R}^{d \times \frac{d}{m}}$
- Let's think about the following computation: $\mathbf{x}\mathbf{W}_i^q$
- **Q1:** What is the number of arithmetic operations? $O(b \times n \times d \times \frac{d}{m})$ for each head
- **Q2:** What about the number of IO? $O(b \times n \times d + d \times \frac{d}{m} + b \times n \times \frac{d}{m})$ for each head
- **Q3:** What are these quantities for all heads?
 - Number of arithmetic ops: $O(bnd^2)$
 - Number of IO ops: $O(2bnd + d^2)$

Computations in Self-Attention Block

b : batch size,
 n : sequence length,
 m : number of heads
 d : feature dimension in output of SA
 d/m : feature dimension inside each SA head
 $d_{\text{ff}} = 4d$: feature dimension inside FFN

Dimensions	Operation	Computations	IO
$\mathbf{x} \in \mathbb{R}^{b \times n \times d}, \mathbf{W}_i^q \in \mathbb{R}^{d \times \frac{d}{m}}$	$\mathbf{x}\mathbf{W}_i^q, \mathbf{x}\mathbf{W}_i^k, \mathbf{x}\mathbf{W}_i^v$ for m heads	$O(bnd^2)$	$O(d^2 + 2bnd)$
$\mathbf{Q}_i, \mathbf{K}_i \in \mathbb{R}^{b \times n \times \frac{d}{m}}$	$P_i \leftarrow \text{softmax}\left(\frac{\mathbf{Q}_i \mathbf{K}_i^T}{\sqrt{d/m}}\right)$ for m heads	$O(bn^2d)$	$O(2bnd + bmn^2)$
$\mathbf{V}_i \in \mathbb{R}^{b \times n \times \frac{d}{m}}, P_i \in \mathbb{R}^{b \times n \times n}$	$\text{head}_i \leftarrow P_i \mathbf{V}_i$ for m heads	$O(bn^2d)$	$O(2bnd + bmn^2)$
$\mathbf{W}^o \in \mathbb{R}^{d \times d}, \text{head}_i \in \mathbb{R}^{b \times n \times \frac{d}{m}}$	$Y = \text{Concat}(\text{head}_1, \dots, \text{head}_m) \mathbf{W}^o$	$O(bnd^2)$	$O(2bnd + d^2)$
$Y \in \mathbb{R}^{b \times n \times d}, \mathbf{W}_1 \in \mathbb{R}^{d \times d_{\text{ff}}}, \mathbf{W}_2 \in \mathbb{R}^{d_{\text{ff}} \times d}$	$Y = \text{ReLU}(Y\mathbf{W}_1)\mathbf{W}_2$	$O(16bnd^2)$	$O(2bnd + 8d^2)$
---	Total	$O(bnd^2 + bn^2d)$	$O(bnd + bmn^2 + d^2)$

The bottlenecks

b : batch size,
 n : sequence length,
 d : feature dimension in output of SA

- So, in total, we have $\rightarrow$
- The quadratic terms are based on n and d
- d is fixed (part of architecture) but n changes with input.
- **Bottlenecks #1:** If n (sequence length) $\gg d$ (feature dimension), the time and space complexity would be dominated by $O(n^2)$.
- **However**, despite this quadratic dependence these are parallelizable operations which can be computed efficiently in GPUs.
 - In comparison, RNNs perform less arithmetic ops but they're not all parallelizable.
- **Bottlenecks #2:** Another potential bottleneck is how fast we can run IO.
(more on this later)

Computations	IO
$O(bnd^2 + bn^2d)$	$O(bnd + bmn^2 + d^2)$

Layer Type	Complexity per Layer	Sequential Operations
Self-Attention	$O(n^2 \cdot d)$	$O(1)$
Recurrent	$O(n \cdot d^2)$	$O(n)$

[[Vaswani et al. 2017](#)]

Recap

- Transformers **computation of a full sequence** is bounded by $O(bnd^2 + bn^2d)$.
 - Generally, the quadratic term that depends on seq len n is more concerning.
- **IO may also impose** other limits on this (coming up).
- Also, the above calculations is for a given sentences (e.g. training time).
 - How bad is the computational complexity during the decoding time where we want to generate text one token at a time?



During decoding time, how slow
is attention computation?



Q: Pulse Check: What is a KV-Cache?

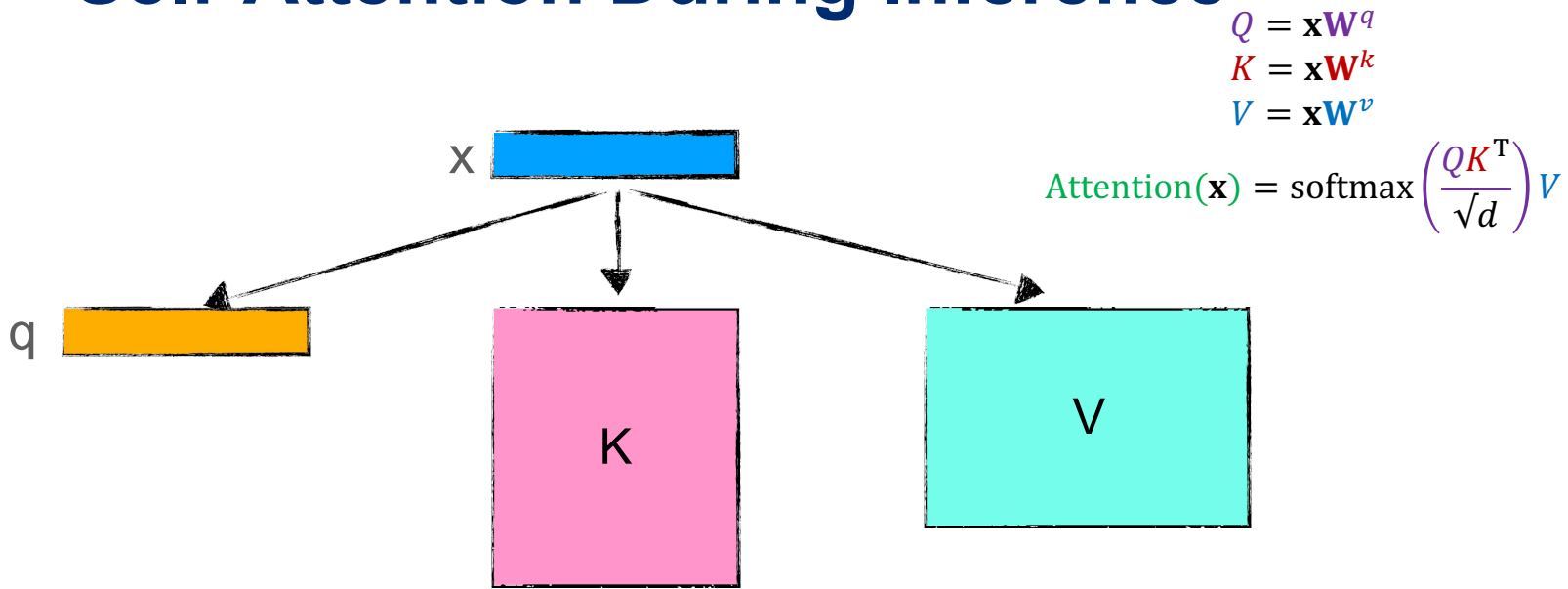
$$Q = \mathbf{x} \mathbf{W}^q$$

$$K = \mathbf{x} \mathbf{W}^k$$

$$V = \mathbf{x} \mathbf{W}^v$$

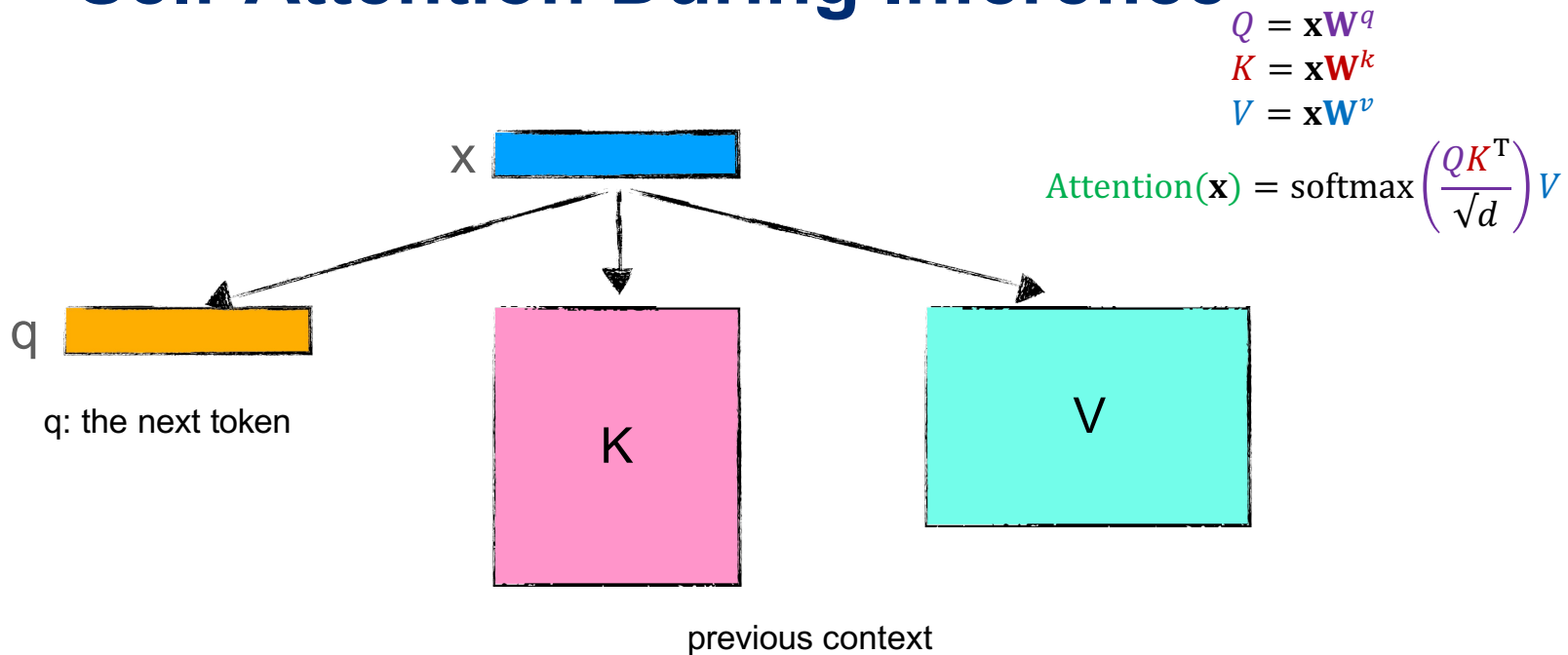
$$\text{Attention}(\mathbf{x}) = \text{softmax} \left(\frac{QK^T}{\sqrt{d}} \right) V$$

Self-Attention During Inference



[Slide credit: Arman Cohan]

Self-Attention During Inference



[Slide credit: Arman Cohan]

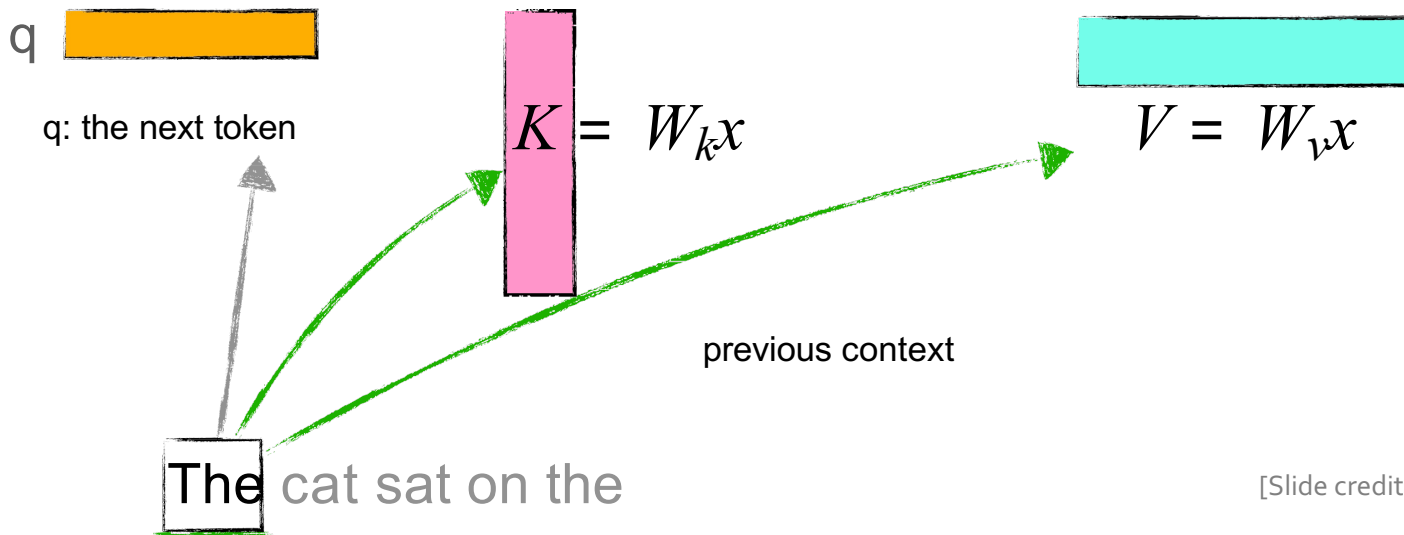
Self-Attention During Inference

$$Q = \mathbf{x} \mathbf{W}^q$$

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[Slide credit: Arman Cohan]

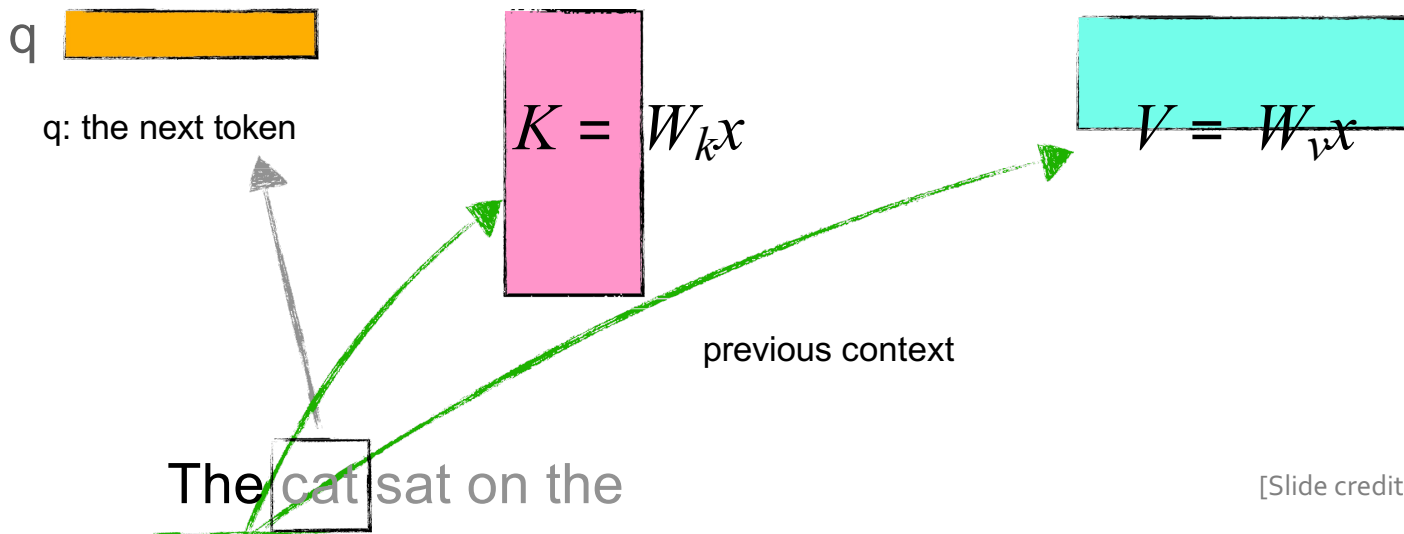
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[Slide credit: Arman Cohan]

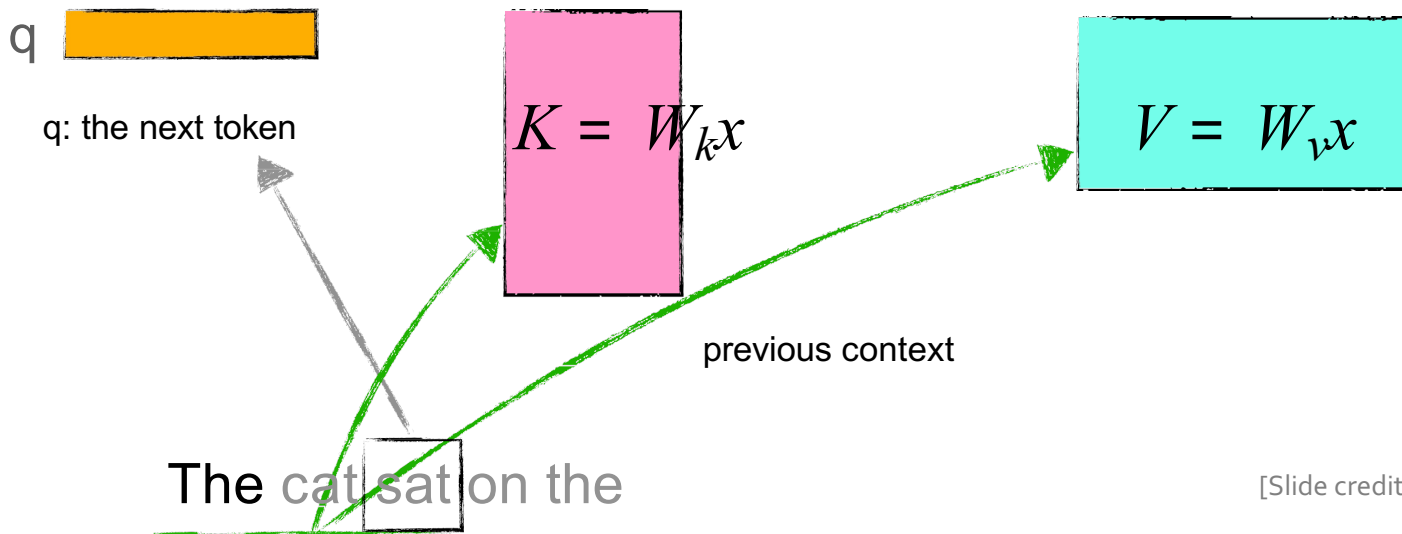
Self-Attention During Inference

$$Q = \mathbf{x}W^q$$

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[Slide credit: Arman Cohan]

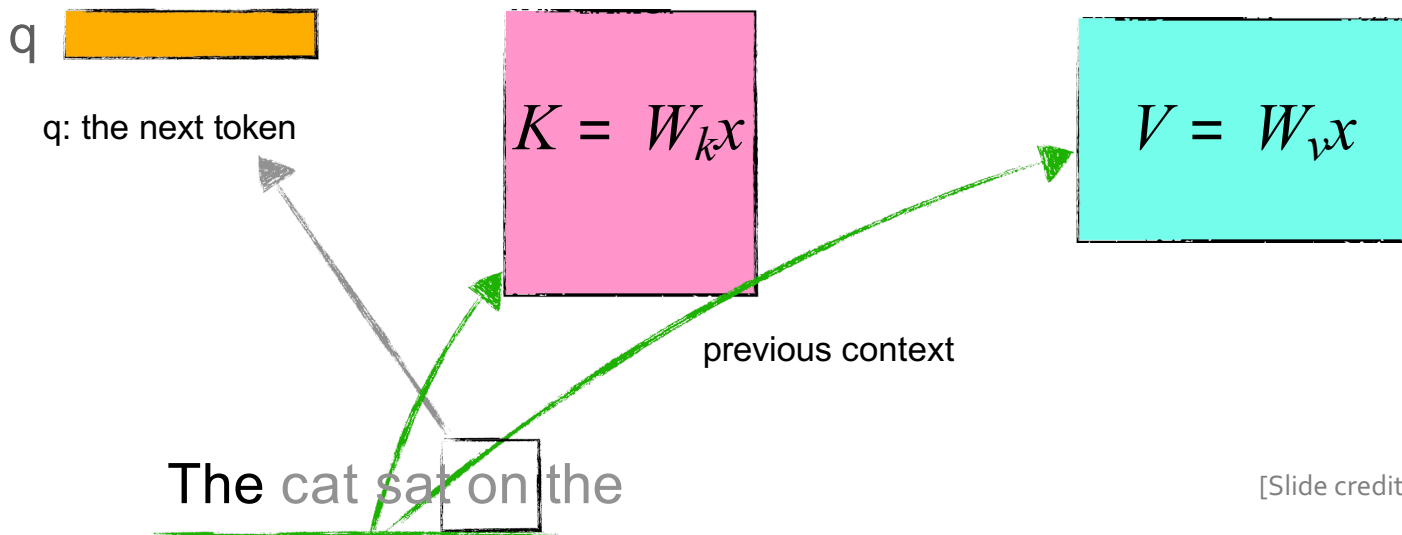
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[Slide credit: Arman Cohan]

Self-Attention During Inference

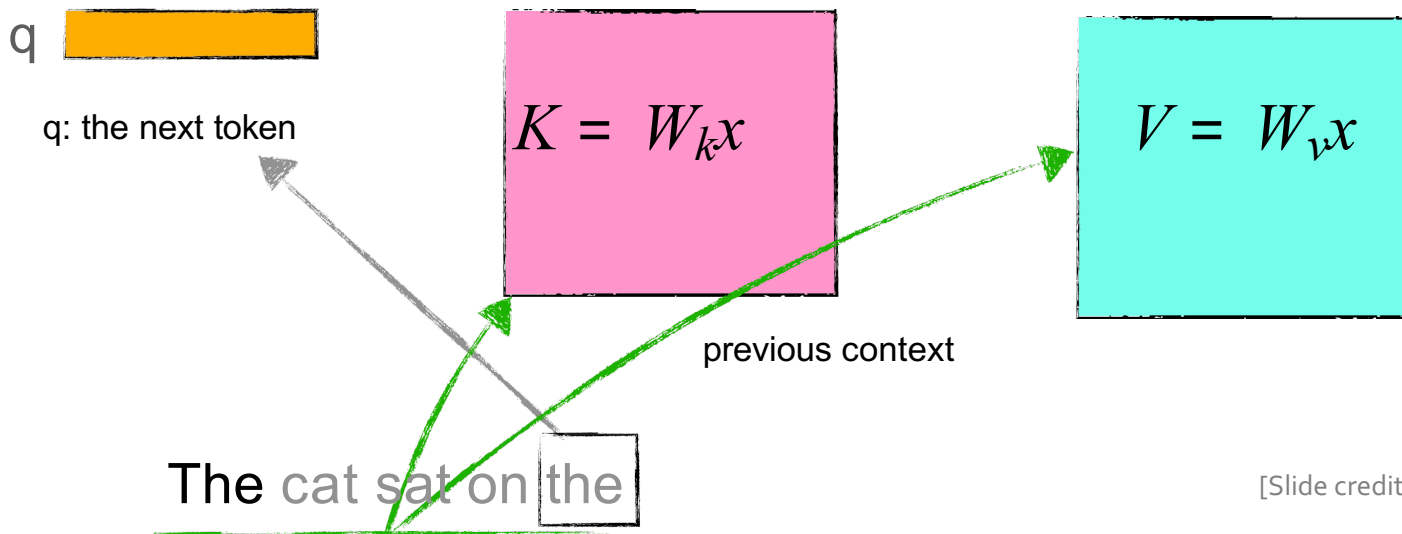
- We are computing the Keys and Values many times!
 - Let's reduce redundancy! 🤔

$$Q = \mathbf{x}W^q$$

$$K = \mathbf{x}W^k$$

$$V = \mathbf{x}W^v$$

$$\text{Attention}(\mathbf{x}) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V$$



[Slide credit: Arman Cohan]

KV-Cache for reducing inference redundancy

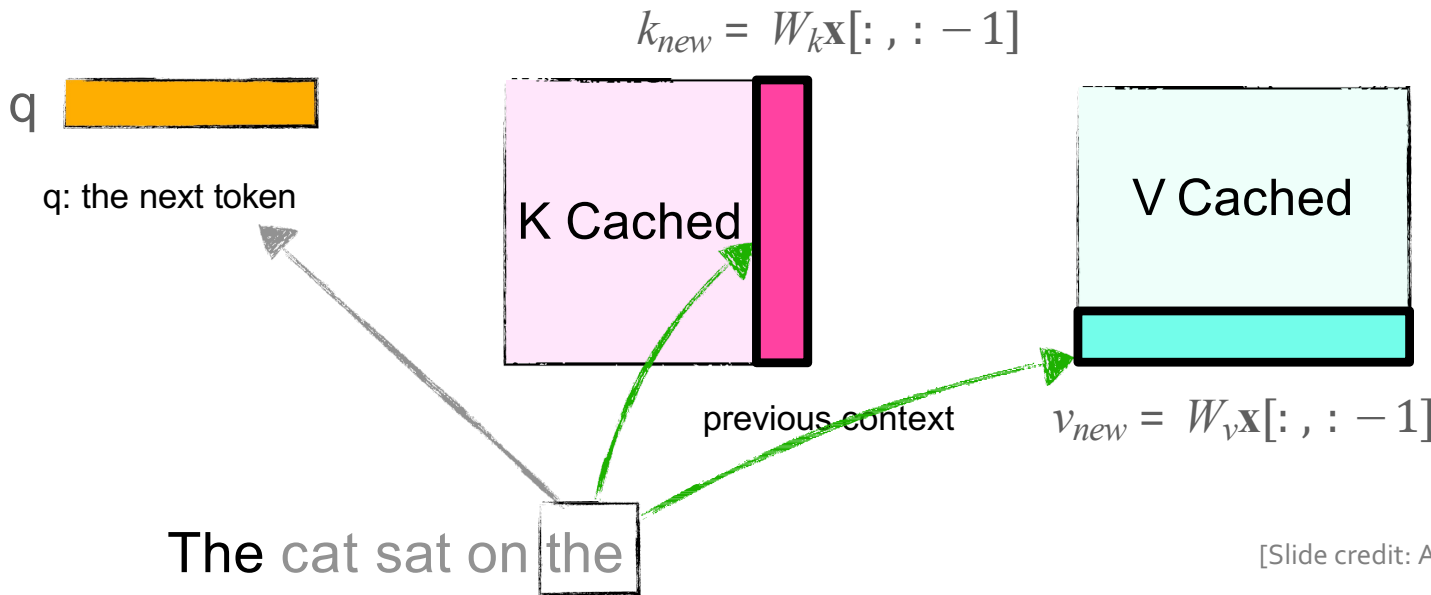
- We are computing the Keys and Values many times!
 - Let's reduce redundancy! 🤔

$$Q = \mathbf{x} \mathbf{W}^q$$

$$K = \mathbf{x} \mathbf{W}^k$$

$$V = \mathbf{x} \mathbf{W}^v$$

$$\text{Attention}(\mathbf{x}) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V$$

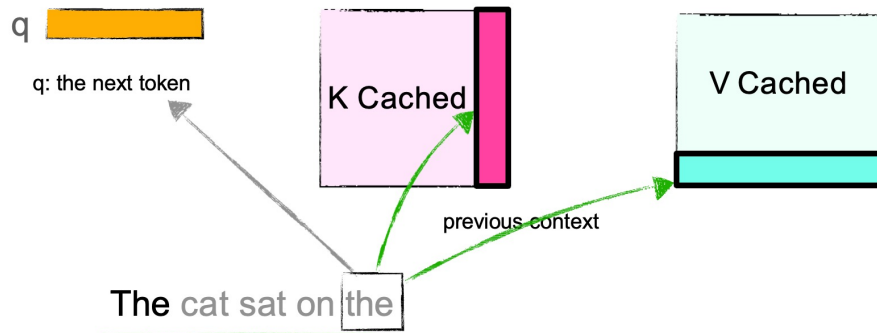


[Slide credit: Arman Cohan]

Quiz: KV-Cache

- How much **memory** does this KV-cache require? Let's assume,
 - batch size b ,
 - embedding dimension is d ,
 - the length of the sequence seen so far is n ,
 - your model has L layers,
 - Each param is stored k bytes (e.g., FP16 takes 2 bytes)

1. $2bnLk$
2. $2bdLk$
3. $bndLk$
4. $2bndLk$



Recap

- To avoid **redundant computations** during decoding time, KV-cache is used to keep track of previous calculations of keys and values.
- But how exactly how costly are these computations?

Decoding Computations (processing token by token)

b : batch size,
 n : sequence length **thus far**,
 m : number of heads
 d : feature dimension in output of SA
 $d_{ff} = 4d$: feature dimension inside FFN

Notice we're doing this computations for one token

Compared to the previous table (SA for a seq of length n), all the cells have one less dependence on n (e.g., $n^2 \rightarrow n$ or $n \rightarrow 1$).

Dimensions	Operation	Computations	IO
$\mathbf{x} \in \mathbb{R}^{b \times 1 \times d}$, $\mathbf{W}_i^q \in \mathbb{R}^{d \times \frac{d}{m}}$	Query/key computations get combined with KV-cache for m heads	$O(bd^2)$	$O(d^2 + 2bd)$
$\mathbf{Q}_i, \mathbf{K}_i \in \mathbb{R}^{b \times 1 \times \frac{d}{m}}$ of the new token; $\mathbf{K}_{kv} \in \mathbb{R}^{b \times n \times \frac{d}{m}}$ from KV-cache	$P_i \leftarrow \text{softmax}\left(\frac{\mathbf{Q}_i [\mathbf{K}_{kv}; \mathbf{K}_i]^T}{\sqrt{d/m}}\right)$ for m heads	$O(bnd)$	$O(bnm + bnd + bd)$
$\mathbf{V}_i \in \mathbb{R}^{b \times 1 \times \frac{d}{m}}$ of the new token; $\mathbf{V}_{kv} \in \mathbb{R}^{b \times n \times \frac{d}{m}}$ from KV-cache; $P_i \in \mathbb{R}^{b \times 1 \times (n+1)}$	$\text{head}_i \leftarrow P_i [\mathbf{V}_{kv}; \mathbf{V}_i]$ for m heads	$O(bnd)$	$O(bnm + bnd + bd)$
$\mathbf{W}^o \in \mathbb{R}^{d \times d}$, $\text{head}_i \in \mathbb{R}^{b \times 1 \times \frac{d}{m}}$	$\mathbf{Y} = \text{Concat}(\text{head}_1, \dots, \text{head}_m) \mathbf{W}^o$	$O(bd^2)$	$O(2bd + d^2)$
$\mathbf{Y} \in \mathbb{R}^{b \times 1 \times d}$, $\mathbf{W}_1 \in \mathbb{R}^{d \times d_{ff}}$, $\mathbf{W}_2 \in \mathbb{R}^{d_{ff} \times d}$	Now the computations (of next token) has linear dependence on seq length.	$O(bd^2)$	$O(2bd + 8d^2)$
---	Total	$O(bd^2 + bnd)$	$O(bmn + bnd + d^2)$

Recap: complexity of decoding text

- Transformers computation for generating each word is $O(bd^2 + bnd)$.
- Remember the complexity of processing a full sequence of length: $O(bnd^2 + bn^2d)$.

Quiz: Enc-Dec inference cost

- Suppose we're using an Enc-Dec model for summarization task. Input are long passages with length m and the expected (summary length is n). The computational complexity of this operation is:
 - $O(n) + O(m)$
 - $O(n) + O(m) + O(nm)$
 - $O(n^2) + O(m^2) + O(nm)$
 - $O(n^2) + O(m^2)$

No, self attention is all-to-all and so quadratic.

Quiz: Enc-Dec inference cost

- Suppose we're using an Enc-Dec model for summarization task. Input are long passages with length m and the expected (summary length is n). The computational complexity of this operation is:
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 - $O(n^2) + O(m^2) + O(nm)$
 - $O(n^2) + O(m^2)$

No, self attention is all-to-all and so quadratic in m and n .

Quiz: Enc-Dec inference cost

- Suppose we're using an Enc-Dec model for summarization task. Input are long passages with length m and the expected (summary length is n). The computational complexity of this operation is:
 - $O(n) + O(m)$
 - $O(n) + O(m) + O(nm)$
 - $O(n^2) + O(m^2) + O(nm)$
 - $O(n^2) + O(m^2)$

No, cross attention is missing.

Quiz: Enc-Dec inference cost

- Suppose we're using an Enc-Dec model for summarization task. Input are long passages with length m and the expected (summary length is n). The computational complexity of this operation is:
 - $O(n) + O(m)$
 - $O(n) + O(m) + O(nm)$
 - $O(n^2) + O(m^2) + O(nm)$
 - $O(n^2) + O(m^2)$

Yes. The three terms are respectively the Encoder self-attention, Decoder self-attention, and Cross attention.

Though a more accurate term is: $O(n^2d + nd^2 + m^2d + md^2 + nmd)$ with d denoting the model embedding dimension.

Summary: Computational Complexity of Transformers

- **Process a sequence at once:** Computation is bounded by $O(n^2)$.
- **Processing one token at a time during inference:**
 - **KV-Cache:** To avoid redundant computations during decoding time, KV-cache is used to keep track of previous calculations of keys and values.
 - The computation is bounded by $O(n)$.
- Though in all cases, the computations are penalizable (modulo Transformer layers).
- **IO bottlenecks:** when reading/writing slows us down; we will get to this (next chapter).

Writing our own Transformer (extra reading offline)

Clone Helper Function

- Create N copies of pytorch nn.Module
- The Transformer's structure contains a lot of design repetition (like VGG)
- Remember these clones shouldn't share parameters (for the most part)

```
def clones(module, N):  
    "Produce N identical layers."  
    return nn.ModuleList([copy.deepcopy(module) for _ in range(N)])
```

Create Embedding

- Create vector representation of sequence vocabulary
- nn.Embedding creates a lookup table to map sequence vocabulary to unique vectors

```
class Embeddings(nn.Module):  
    def __init__(self, d_model, vocab):  
        super(Embeddings, self).__init__()  
        self.lut = nn.Embedding(vocab, d_model)  
        self.d_model = d_model  
  
    def forward(self, x):  
        return self.lut(x) * math.sqrt(self.d_model)
```

Positional Encoding

- Add information about an element's position in a sequence to its representation
- Element wise addition of sinusoidal encoding



```
class PositionalEncoding(nn.Module):
    "Implement the PE function."

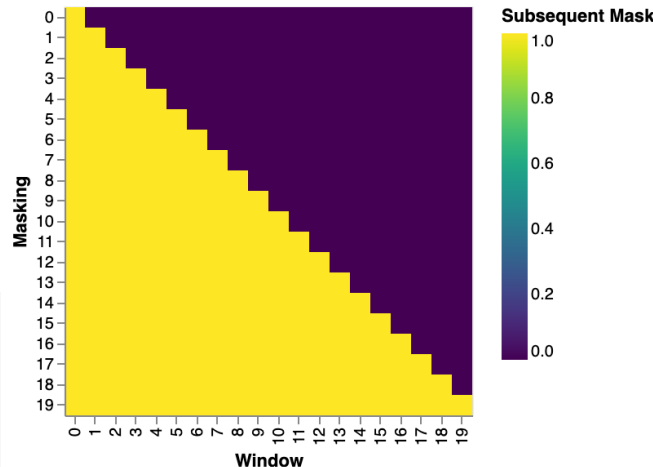
    def __init__(self, d_model, dropout, max_len=5000):
        super(PositionalEncoding, self).__init__()
        self.dropout = nn.Dropout(p=dropout)

        # Compute the positional encodings once in log space.
        pe = torch.zeros(max_len, d_model)
        position = torch.arange(0, max_len).unsqueeze(1)
        div_term = torch.exp(
            torch.arange(0, d_model, 2) * -(math.log(10000.0) / d_model)
        )
        pe[:, 0::2] = torch.sin(position * div_term)
        pe[:, 1::2] = torch.cos(position * div_term)
        pe = pe.unsqueeze(0)
        self.register_buffer("pe", pe)

    def forward(self, x):
        x = x + self.pe[:, : x.size(1)].requires_grad_(False)
        return self.dropout(x)
```

Attention block

```
def attention(query, key, value, mask=None, dropout=None):  
    "Compute 'Scaled Dot Product Attention'"  
    d_k = query.size(-1)  
    scores = torch.matmul(query, key.transpose(-2, -1)) / math.sqrt(d_k)  
    if mask is not None:  
        scores = scores.masked_fill(mask == 0, -1e9)  
    p_attn = scores.softmax(dim=-1)  
    if dropout is not None:  
        p_attn = dropout(p_attn)  
    return torch.matmul(p_attn, value), p_attn
```



$-1e9$ is a large negative number,
which leads to $\text{softmax}(-1e9) \approx 0$

Multi-Head Attention

```
class MultiHeadedAttention(nn.Module):
    def __init__(self, h, d_model, dropout=0.1):
        "Take in model size and number of heads."
        super(MultiHeadedAttention, self).__init__()
        assert d_model % h == 0
        # We assume d_v always equals d_k
        self.d_k = d_model // h
        self.h = h
        self.linears = clones(nn.Linear(d_model, d_model), 4)
        self.attn = None
        self.dropout = nn.Dropout(p=dropout)
```

```
def forward(self, query, key, value, mask=None):
    "Implements Figure 2"
    if mask is not None:
        # Same mask applied to all h heads.
        mask = mask.unsqueeze(1)
    nbatches = query.size(0)

    # 1) Do all the linear projections in batch from d_model => h x d_k
    query, key, value = [
        lin(x).view(nbatches, -1, self.h, self.d_k).transpose(1, 2)
        for lin, x in zip(self.linears, (query, key, value))
    ]

    # 2) Apply attention on all the projected vectors in batch.
    x, self.attn = attention(
        query, key, value, mask=mask, dropout=self.dropout
    )

    # 3) "Concat" using a view and apply a final linear.
    x = (
        x.transpose(1, 2)
        .contiguous()
        .view(nbatches, -1, self.h * self.d_k)
    )
    del query
    del key
    del value
    return self.linears[-1](x)
```

FeedForward Layer

```
class PositionwiseFeedForward(nn.Module):  
    "Implements FFN equation."  
  
    def __init__(self, d_model, d_ff, dropout=0.1):  
        super(PositionwiseFeedForward, self).__init__()  
        self.w_1 = nn.Linear(d_model, d_ff)  
        self.w_2 = nn.Linear(d_ff, d_model)  
        self.dropout = nn.Dropout(dropout)  
  
    def forward(self, x):  
        return self.w_2(self.dropout(self.w_1(x).relu()))
```

Sublayer Connections

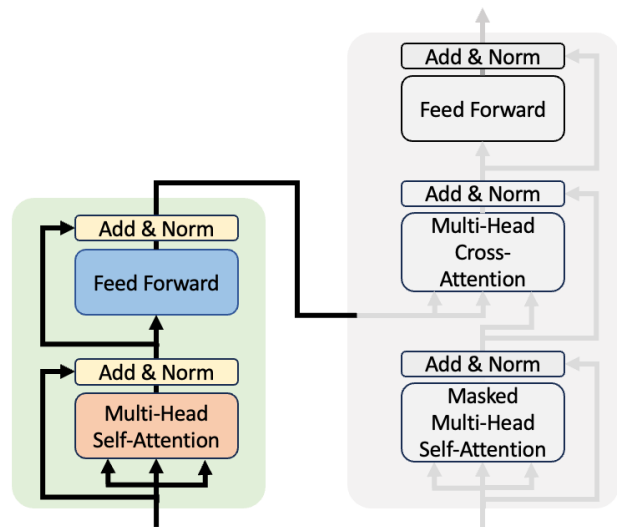
```
class SublayerConnection(nn.Module):
    """
    A residual connection followed by a layer norm.
    Note for code simplicity the norm is first as opposed to last.
    """

    def __init__(self, size, dropout):
        super(SublayerConnection, self).__init__()
        self.norm = LayerNorm(size)
        self.dropout = nn.Dropout(dropout)

    def forward(self, x, sublayer):
        "Apply residual connection to any sublayer with the same size."
        return x + self.dropout(sublayer(self.norm(x)))
```

Encoder Layer

```
class EncoderLayer(nn.Module):  
    "Encoder is made up of self-attn and feed forward (defined below)"  
  
    def __init__(self, size, self_attn, feed_forward, dropout):  
        super(EncoderLayer, self).__init__()  
        self.self_attn = self_attn  
        self.feed_forward = feed_forward  
        self.sublayer = clones(SublayerConnection(size, dropout), 2)  
        self.size = size  
  
    def forward(self, x, mask):  
        "Follow Figure 1 (left) for connections."  
        x = self.sublayer[0](x, lambda x: self.self_attn(x, x, x, mask))  
        return self.sublayer[1](x, self.feed_forward)
```



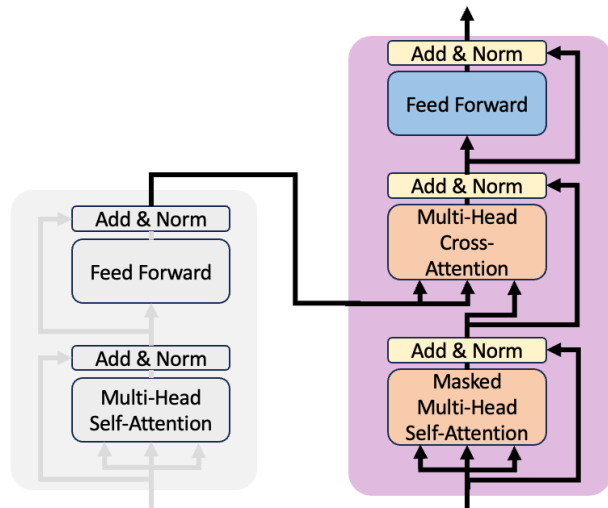
Decoder Layer

- Same as encoder layers other than:
 - the additional multi-head attention block to preform cross-attention with the output representation from the encoder

```
class DecoderLayer(nn.Module):
    "Decoder is made of self-attn, src-attn, and feed forward (defined below)"

    def __init__(self, size, self_attn, src_attn, feed_forward, dropout):
        super(DecoderLayer, self).__init__()
        self.size = size
        self.self_attn = self_attn
        self.src_attn = src_attn
        self.feed_forward = feed_forward
        self.sublayer = clones(SublayerConnection(size, dropout), 3)

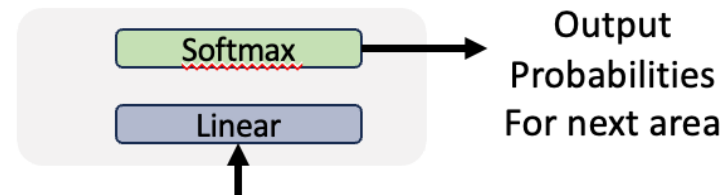
    def forward(self, x, memory, src_mask, tgt_mask):
        "Follow Figure 1 (right) for connections."
        m = memory
        x = self.sublayer[0](x, lambda x: self.self_attn(x, x, x, tgt_mask))
        x = self.sublayer[1](x, lambda x: self.src_attn(x, m, m, src_mask))
        return self.sublayer[2](x, self.feed_forward)
```



The Prediction Head

- A final linear mapping
- Apply softmax to convert logits to probabilities

```
class Generator(nn.Module):  
    "Define standard linear + softmax generation step."  
  
    def __init__(self, d_model, vocab):  
        super(Generator, self).__init__()  
        self.proj = nn.Linear(d_model, vocab)  
  
    def forward(self, x):  
        return log_softmax(self.proj(x), dim=-1)
```



Build each block

```
class Encoder(nn.Module):
    "Core encoder is a stack of N layers"

    def __init__(self, layer, N):
        super(Encoder, self).__init__()
        self.layers = clones(layer, N)
        self.norm = LayerNorm(layer.size)

    def forward(self, x, mask):
        "Pass the input (and mask) through each layer in turn."
        for layer in self.layers:
            x = layer(x, mask)
        return self.norm(x)
```

```
class Decoder(nn.Module):
    "Generic N layer decoder with masking."

    def __init__(self, layer, N):
        super(Decoder, self).__init__()
        self.layers = clones(layer, N)
        self.norm = LayerNorm(layer.size)

    def forward(self, x, memory, src_mask, tgt_mask):
        for layer in self.layers:
            x = layer(x, memory, src_mask, tgt_mask)
        return self.norm(x)
```

```

class EncoderDecoder(nn.Module):
    """
    A standard Encoder-Decoder architecture. Base for this and many
    other models.
    """

    def __init__(self, encoder, decoder, src_embed, tgt_embed, generator):
        super(EncoderDecoder, self).__init__()
        self.encoder = encoder
        self.decoder = decoder
        self.src_embed = src_embed
        self.tgt_embed = tgt_embed
        self.generator = generator

    def forward(self, src, tgt, src_mask, tgt_mask):
        "Take in and process masked src and target sequences."
        return self.decode(self.encode(src, src_mask), src_mask, tgt, tgt_mask)

    def encode(self, src, src_mask):
        return self.encoder(self.src_embed(src), src_mask)

    def decode(self, memory, src_mask, tgt, tgt_mask):
        return self.decoder(self.tgt_embed(tgt), memory, src_mask, tgt_mask)

```

```

def make_model(
    src_vocab, tgt_vocab, N=6, d_model=512, d_ff=2048, h=8, dropout=0.1
):
    "Helper: Construct a model from hyperparameters."
    c = copy.deepcopy
    attn = MultiHeadedAttention(h, d_model)
    ff = PositionwiseFeedForward(d_model, d_ff, dropout)
    position = PositionalEncoding(d_model, dropout)
    model = EncoderDecoder(
        Encoder(EncoderLayer(d_model, c(attn), c(ff), dropout), N),
        Decoder(DecoderLayer(d_model, c(attn), c(attn), c(ff), dropout), N),
        nn.Sequential(Embeddings(d_model, src_vocab), c(position)),
        nn.Sequential(Embeddings(d_model, tgt_vocab), c(position)),
        Generator(d_model, tgt_vocab),
    )

    # This was important from their code.
    # Initialize parameters with Glorot / fan_avg.
    for p in model.parameters():
        if p.dim() > 1:
            nn.init.xavier_uniform_(p)
    return model

```